
Diffusion on Hyperbolic Space with a Curvature Determined Confinement

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Abstract

Score based generative models need a forward stochastic process whose law converges to a reference distribution that can be sampled. Hyperbolic space $\mathbb{H}^n$ is a natural state space for hierarchical data, but Brownian motion on it is transient and has no invariant probability law. We analyse the confined diffusion $dX_t = -\frac{c}{2}\nabla r^2 dt + \sqrt{2} dB_t$, whose invariant law $\mu_c \propto e^{-cr^2/2} d\text{vol}$ exists for every $c > 0$, and trace its spectral structure into the design, training and validation of a generative model. The spectral gap $\lambda_1(c)$ sets the rate at which the forward process forgets the data, and the curvature threshold $c = n-1$ separates two regimes. Above the threshold the gap grows linearly, with $c-(n-1) \leq \lambda_1(c) \leq c-\frac{2}{3}(n-1)+o(1)$. Below it the gap collapses faster than any power of c as $c \downarrow 0$, while an explicit lower bound keeps it positive at every c . On $\mathbb{H}^2$ the upper and lower bounds match: $\lambda_1(c) \asymp \sqrt{c} e^{-1/(2c)}$. A commutation identity shows that for every c the gap is attained in the first spherical harmonic sector, with the slowest radial mode relaxing exactly c faster than the slowest angular mode. The first angular harmonic of the data is therefore the last component the forward process forgets, and the sampler must reconstruct it from the least informative endpoint. The sampler’s score is the stationary drift plus a learned residual read in an orthonormal frame, which avoids the $\Theta(e^r)$ amplification that makes untrained extrinsic network heads diverge in a one seed ablation. Simulated eigenfunction observables relax at the computed gaps to within four percent in dimensions two to four. On hierarchical data in $\mathbb{H}^2$ and $\mathbb{H}^3$, synthetic and built on the WordNet mammal taxonomy, the generation error at a fixed horizon is smallest at or within a factor two of the threshold and rises on either side of it, for different reasons. Below the threshold the forward process does not reach its prior, which a statistic of the forward process alone predicts before training. Above it the rise is consistent with a harder score estimation and sampling problem. Across repeated training seeds the intrinsic model is comparable to a wrapped normal baseline trained with exact score targets.

1 Introduction

Denosing diffusion and score based models generate data by reversing a stochastic process: the forward process carries the data law towards a reference law, and a learned reverse process transports it back [1, 2]. On $\mathbb{R}^n$ the standard choice is a variance preserving Ornstein–Uhlenbeck (OU)

process with a Gaussian reference law [2]. On compact manifolds, Brownian motion already has the normalised volume as a stationary law, which is used for spherical and rotational data [3], for torsion angles on tori [5] and, combined with Euclidean translational noising, for rigid body and protein geometry [6, 7]; E(3)-equivariant Euclidean diffusion is also used for three-dimensional molecular coordinates [8]. Discrete state spaces use Markov corruption processes [9, 10].

The normalised volume, the reference law on compact manifolds, does not exist on hyperbolic space $\mathbb{H}^n$, a natural state space for hierarchical data: trees admit embeddings in $\mathbb{H}^2$ with arbitrarily low distortion [13], while hyperbolic embeddings are effective for learned hierarchical representations [12]. The space $\mathbb{H}^n$ has infinite volume, Brownian motion on it is transient [30], and the bottom of the spectrum of $-\Delta$ is $(n-1)^2/4$ [18, 28]; undrifted Brownian motion therefore has no stationary probability law. Alternatives are the wrapped normal, a Euclidean Gaussian pushed through the exponential map [11], and Riemannian diffusion models with a drift towards a prescribed prior [3, 4]. We study radial quadratic confinement, $W_c = \frac{c}{2}r^2$ with generator $\mathcal{L}_c = \Delta - \nabla W_c \cdot \nabla$ and invariant law $\mu_c \propto e^{-cr^2/2} d\text{vol}$, and ask when the stationary law exists, how quickly the process mixes and how those rates affect a generative model.

The paper follows the confined diffusion from its stochastic analysis to a sampler, organised around two questions: how the invariant law and the spectral gap of the forward process guide the choice of c and supply a validation statistic available before any network is trained, and how the stationary dynamics can be built into the reverse process so that the sampler stays stable outside the training support. Our contributions are as follows.

- **Spectral estimates on both sides of the threshold** (Section 3). The law μ_c exists and is the unique invariant probability measure for every $c > 0$; above the threshold $c = n - 1$ the Bakry–Émery bound gives $\lambda_1(c) \geq c - (n - 1)$ and a Rayleigh quotient an upper bound of the same order (Theorems 3.1, 3.2). A commutation identity (Proposition 3.3) shows that the nonzero radial spectrum of $-\mathcal{L}_c$ is the first spherical harmonic spectrum shifted by exactly c : the gap is attained in the first harmonic sector for every c , the slowest radial mode relaxes c faster than the slowest angular mode, and $\lambda_1(c) \leq \text{Var}_{\mu_c}(r)^{-1} - c$ (Corollary 3.4). Theorem 3.5 sharpens this bound at large c to $\lambda_1(c) \leq c - \frac{2}{3}(n - 1) + o(1)$, hence $\lambda_1(c)/c \rightarrow 1$. Below the threshold the gap instead collapses exponentially (Theorem 3.6), and a positive lower bound with explicit constants (Theorem 3.7) matches the collapse on $\mathbb{H}^2$, where $\lambda_1(c) \asymp \sqrt{c} e^{-1/(2c)}$, and its exponent on $\mathbb{H}^3$. Since the gap stays positive for every c (Remark 3.8), the threshold is a crossover in the mixing rate, not a boundary of existence.
- **A stable score model** (Section 4). The score is the stationary drift plus a learned residual in an orthonormal frame, so a zero residual recovers the ergodic stationary process exactly, and the frame avoids the $\Theta(e^r)$ amplification of extrinsic outputs under the Minkowski projection (Lemma B.1), which otherwise creates a feedback loop once a sample leaves the training support.
- **Experiments that validate the theory and resolve the two regimes** (Section 5). Simulated eigenfunction observables on $\mathbb{H}^2$, $\mathbb{H}^3$ and $\mathbb{H}^4$ relax at the finite element spectral gaps to within four percent, and the slowest radial and angular modes differ in rate by the amount c that Proposition 3.3 predicts. On hierarchical data in $\mathbb{H}^2$ and $\mathbb{H}^3$, including one built on the WordNet mammal taxonomy, the generation error at a fixed horizon is smallest at or within a factor two of the threshold and rises on both sides. Below the threshold the failure is visible in the forward process before training. Across repeated training seeds the intrinsic model is comparable to a wrapped baseline with exact targets. Section 2 and Appendix A place $\mathbb{H}^n$ among the geometries in use: finite volume, not compactness, decides whether Brownian motion needs a drift.

Related work. Riemannian diffusion models [3, 4] develop forward and reverse diffusions and trainable objectives on manifolds, including synthetic non-compact hyperbolic examples with a drift towards a chosen prior. Our question is more specific: we analyse radial quadratic confinement on $\mathbb{H}^n$, identify its curvature dependent spectral regimes and trace them into score parameterisation and generative failure. Wrapped normal models [11] and hyperbolic representation learning [12, 13] motivate the geometry; the spectral analysis uses Bakry–Émery theory [32], Lyapunov type

Poincaré criteria [19] and the polar-decomposition isospectrality of A^*A and AA^* [21], with Deift's commutation method providing a related spectral precedent [20]. We do not claim that radial quadratic confinement is the optimal generator; our aim is to characterise its geometric cost.

2 Brownian motion, invariant measures and the need for a drift

Let (M, g) be a smooth connected Riemannian manifold without boundary, with volume vol_g and Laplace–Beltrami operator Δ (unit Laplacian convention), so that Brownian motion has generator Δ and, when conservative, semigroup $P_t = e^{t\Delta}$ [30]. A probability measure is invariant if P_t preserves it; vol_g is always infinitesimally invariant (Proposition A.1), so the question is whether it can be normalised. Theorem A.2 (Appendix A) answers this question for conservative diffusions. If $\text{vol}_g(M) < \infty$, the normalised volume $\bar{\mu}$ is the unique invariant probability measure of (X_t) , and on a compact M the process mixes exponentially at the spectral gap of $-\Delta$. If $\text{vol}_g(M) = \infty$, no invariant probability measure has constant density. A drift can supply one: for $e^{-V} \in L^1(\text{vol}_g)$ the law $d\mu_V \propto e^{-V} d\text{vol}_g$ is the unique invariant probability law of the diffusion generated by $\mathcal{L}_V = \Delta - \nabla V \cdot \nabla$.

The theorem separates two roles often assigned to compactness: finite volume normalises the volume measure, whereas compactness supplies a discrete spectrum and a positive gap; a hyperbolic surface of finite area that is not compact has the normalised volume as stationary law but essential spectrum $[1/4, \infty)$ [29] (Proposition A.3). Because $\mathbb{H}^n$ is homogeneous with infinite volume, Corollary A.4(c) forces a drift. Appendix A records the rest of the classification, and the remainder of the paper quantifies what the drift costs on $\mathbb{H}^n$.

3 The hyperbolic Gaussian and its spectral gap

We work on $\mathbb{H}^n$, $n \geq 2$, in the hyperboloid model $\{x \in \mathbb{R}^{n+1} : \langle x, x \rangle_L = -1, x_{n+1} > 0\}$, where $\langle x, y \rangle_L = \sum_{i \leq n} x_i y_i - x_{n+1} y_{n+1}$ and the sectional curvature is -1 . We take $o = (0, \dots, 0, 1)$ as the base point, write $r(x) = d(o, x)$ and, for $c > 0$, set

$$W_c = \frac{c}{2} r^2, \quad \mathcal{L}_c = \Delta - \nabla W_c \cdot \nabla, \quad dX_t = -\nabla W_c(X_t) dt + \sqrt{2} dB_t,$$

with unnormalised invariant density e^{-W_c} with respect to $d\text{vol}$. Write $\lambda_1(c)$ for the spectral gap of $-\mathcal{L}_c$ in $L^2(\mu_c)$, equivalently the optimal Poincaré constant

$$\lambda_1(c) = \inf \left\{ \frac{\int |\nabla f|^2 d\mu_c}{\text{Var}_{\mu_c}(f)} : f \text{ smooth, bounded, nonconstant} \right\}.$$

In geodesic polar coordinates about o , $d\mu_c = \rho_c(s) ds d\sigma(\theta)$ with $\rho_c(s) = Z_c^{-1} \sinh^{n-1}(s) e^{-cs^2/2}$ and σ the normalised measure on S^{n-1} .

3.1 The confined diffusion exists for every $c > 0$

Theorem 3.1 (A hyperbolic Gaussian whose gap is set by curvature). *On $\mathbb{H}^n$, for $\mathcal{L}_c$ as above:*

- (i) *for every $c > 0$, $d\mu_c \propto e^{-\frac{c}{2}r^2} d\text{vol}$ is a probability measure and the unique invariant law of the diffusion, which is non-explosive;*
- (ii) *if $c > n - 1$, then $\text{Ric} + \nabla^2 W_c \geq (c - (n - 1))g$ and $\mathcal{L}_c$ satisfies the Bakry–Émery condition $\text{CD}(c - (n - 1), \infty)$; hence $\lambda_1(c) \geq c - (n - 1)$ and*

$$\|P_t f - \mu_c(f)\|_{L^2(\mu_c)} \leq e^{-(c-(n-1))t} \|f - \mu_c(f)\|_{L^2(\mu_c)}.$$

The proof (Appendix B) sets the Gaussian decay of $e^{-cr^2/2}$ against the volume growth $\sinh^{n-1} r$ and uses the Hessian comparison $\nabla^2(\frac{1}{2}r^2) \geq g$. The threshold $c = n - 1$ is the magnitude of the negative Ricci term, above which the convexity of the potential dominates it; for curvature $-a^2$ the condition is $c > (n - 1)a^2$, and $a \rightarrow 0$ recovers the Euclidean OU process.

3.2 An upper bound of matching order

Theorem 3.2 (An upper bound of matching order). *For every $c > 0$,*

$$\lambda_1(c) \leq \frac{1}{\text{Var}_{\mu_c}(r)}.$$

Here $c \text{Var}_{\mu_c}(r) \rightarrow v_n$ as $c \rightarrow \infty$, where $v_n = n - 2\left(\Gamma(\frac{n+1}{2})/\Gamma(\frac{n}{2})\right)^2 \in (0, 1]$ is the variance of a χ_n random variable. Consequently there is $c_0(n)$ with

$$c - (n - 1) \leq \lambda_1(c) \leq \frac{2}{v_n} c \quad \text{for all } c \geq c_0(n).$$

The spectral gap therefore grows linearly in the confinement and Theorem 3.1(ii) is order sharp.

The first inequality tests the Rayleigh quotient with the 1-Lipschitz function $f = r$; for large c the radial law approaches a χ_n law at scale $c^{-1/2}$, so the bound has the Euclidean scaling of the OU gap c .

3.3 The gap sits in the first spherical harmonic sector

Since $\mathcal{L}_c$ commutes with the rotations fixing o , $L^2(\mu_c)$ decomposes into sectors indexed by the degree $\ell \geq 0$ of spherical harmonics on S^{n-1} , and on the sector spanned by $g(r)Y(\theta)$ with Y of degree ℓ the operator $-\mathcal{L}_c$ acts on the radial factor as

$$-\mathcal{L}_c^{(\ell)} = -\partial_s^2 - a(s)\partial_s + \frac{\ell(\ell + n - 2)}{\sinh^2 s}, \quad a(s) := (\log \rho_c)'(s) = (n - 1) \coth s - cs, \quad (1)$$

with Dirichlet form $\mathcal{E}_\ell(g) = \int_0^\infty (g'^2 + \ell(\ell + n - 2) \sinh^{-2}(s) g^2) \rho_c ds$ on $\mathcal{H} := L^2((0, \infty), \rho_c ds)$. Write $\lambda^{(\ell)}(c)$ for the bottom of the spectrum of $-\mathcal{L}_c^{(\ell)}$, excluding the eigenvalue 0 of the constants when $\ell = 0$. The following identity is the hyperbolic analogue of the OU ladder, where ∂_{x_1} maps the radial eigenfunction $|x|^2 - n/c$ to x_1 and lowers the eigenvalue from $2c$ to c .

Proposition 3.3 (Radial and angular spectra differ by exactly c). *Let $n \geq 2$ and $c > 0$.*

(i) *The nonzero spectrum of $-\mathcal{L}_c^{(0)}$ equals the spectrum of $-\mathcal{L}_c^{(1)}$ shifted by c , with multiplicity. In particular $\lambda^{(0)}(c) = \lambda^{(1)}(c) + c$, and if F is a radial eigenfunction with eigenvalue $\Lambda > 0$, then $F'(r)\theta_1$ is an eigenfunction in the degree one sector with eigenvalue $\Lambda - c$.*

(ii) *$\lambda^{(\ell)}(c) \geq \lambda^{(1)}(c)$ for every $\ell \geq 1$.*

(iii) *Consequently $\lambda_1(c) = \lambda^{(1)}(c) = \lambda^{(0)}(c) - c$ for every $c > 0$: the spectral gap is attained in the first spherical harmonic sector; and the slowest radial mode relaxes exactly c faster than the slowest angular mode.*

The proof (Appendix B.3, which also proves the corollary below) writes $-\mathcal{L}_c^{(0)} = D^*D$ and $-\mathcal{L}_c^{(1)} + c = DD^*$ for the closed operator $D = \partial_s$ on $\mathcal{H}$ and uses the polar-decomposition isospectrality of D^*D and DD^* off zero [21]; the shift is c because $-a'(s) = (n - 1) \sinh^{-2} s + c$, the angular potential of the first harmonic plus the confinement. Three consequences matter for the rest of the paper.

Corollary 3.4 (Sharpened upper bound and the slow mode). *For every $n \geq 2$ and $c > 0$,*

$$\lambda_1(c) \leq \frac{1}{\text{Var}_{\mu_c}(r)} - c, \quad \text{and} \quad 1 \leq \liminf_{c \rightarrow \infty} \frac{\lambda_1(c)}{c} \leq \limsup_{c \rightarrow \infty} \frac{\lambda_1(c)}{c} \leq \frac{1}{v_n} - 1.$$

Moreover, if the initial law has a density $p_0/\mu_c \in L^2(\mu_c)$ with a nonzero component along $g_1(r)\theta_1$, for g_1 the ground state of the degree one sector, then $\mathbb{E}\theta_1(X_t)$ decays at the asymptotic rate $\lambda_1(c)$, whereas $\mathbb{E}f(X_t) - \mu_c(f)$ decays at least at rate $\lambda_1(c) + c$ for every radial $f \in L^2(\mu_c)$.

Table 5 in Appendix C reports $\lambda_1(c)$ on $\mathbb{H}^2$ computed from (1) by a linear finite element method, and Table 6 there does the same on $\mathbb{H}^3$, $\mathbb{H}^4$ and $\mathbb{H}^5$: the identity $\lambda^{(0)} = \lambda^{(1)} + c$ holds to discretisation error throughout, the sharpened bound tightens as c decreases below the threshold, and in Section 5.1 simulated eigenfunction observables relax at the computed gaps to within four percent.

Theorem 3.5 (The leading order at strong confinement). *For $n \geq 2$ and every $c > 0$,*

$$\lambda_1(c) \leq 1 + \frac{n}{\mathbb{E}_{\mu_c}[\sinh^2 r]} = c - \frac{2}{3}(n-1) + o(1) \quad (c \rightarrow \infty).$$

With Theorem 3.1(ii), $-(n-1) \leq \liminf_{c \rightarrow \infty} (\lambda_1(c) - c) \leq \limsup_{c \rightarrow \infty} (\lambda_1(c) - c) \leq -\frac{2}{3}(n-1)$, so $\lambda_1(c) = c + O(1)$ and $\lambda_1(c)/c \rightarrow 1$.

The test function is the ambient coordinate $x_1 = \sinh(r) \theta_1$ (Appendix B.4), whose Rayleigh quotient reproduces the leading term of the formal expansion of Appendix B.9, confirmed numerically for $n = 2, \dots, 5$ (Table 7, Remark B.2). Since $\lambda_1(c)/c \rightarrow 1$, the upper end $v_n^{-1} - 1$ of the ratio interval in Corollary 3.4 is not attained.

3.4 The collapse below the threshold

Theorem 3.6 (Collapse below the threshold). *There are constants $C(n) < \infty$ such that for all $0 < c \leq 1$: (i) on $\mathbb{H}^2$, $\lambda_1(c) \leq C \sqrt{c} e^{-1/(2c)}$; (ii) on $\mathbb{H}^n$, $n \geq 3$, $\lambda_1(c) \leq C(n) e^{-\beta(n)/c}$ with $\beta(n) = \min \left\{ \frac{(n-1)^2}{2}, 2(n-2) \right\}$. In particular $\lambda_1(c) \rightarrow 0$ faster than any power of c as $c \downarrow 0$.*

Slow angular modes drive the bound. Below the threshold most equilibrium mass lies in a shell near $r_* \approx (n-1)/c$, far from the origin. For a first spherical harmonic cut off near the origin, the cutoff therefore costs only the exponentially small mass near o , and the angular Dirichlet energy is comparable to $\mathbb{E}_{\mu_c}[\sinh^{-2} r; r \geq 1]$, of order $e^{-2(n-2)/c}$ for $n \geq 3$. These two costs give the exponents in $\beta(n)$, and the angular one, $2(n-2)$, is no larger for $n \geq 3$. On $\mathbb{H}^2$ both costs are of order $Z_c^{-1} \asymp \sqrt{c} e^{-1/(2c)}$. Proposition 3.3 explains why the collapse is angular: $-(\log \rho_c)'' = c + (n-1) \sinh^{-2} s \geq c$, so the radial gap $\lambda^{(0)}(c) = c + \lambda_1(c)$ is at least c for every c while the angular gap is exponentially small. The exponents are consistent with being sharp: least squares fits of numerically computed $\log \lambda_1(c)$ against $1/c$ on $\mathbb{H}^n$, $n = 2, \dots, 5$ (Appendix C), give 0.48, 1.97, 3.94 and 5.98 against the predicted $\frac{1}{2}, 2, 4$ and 6. The lower bound follows.

Theorem 3.7 (Lower bounds for every confinement). *Let $n \geq 2$ and $c > 0$, let $w = Z_c \rho_c$ and $W_B = \int_1^\infty w$, let κ_c be the infimum of $\int_0^1 (\eta'^2 + (n-1)\eta^2 / \sinh^2 s) w ds$ over η with $\eta(1) = 1$, and let $D_c = W_B^{-2} \int_1^\infty (\int_s^\infty w)^2 w(s)^{-1} ds$. Then*

$$(i) \quad \lambda_1(c) \geq \min \left\{ \frac{c}{2}, \frac{n-1}{2 \sinh^2 1}, \frac{\min(\kappa_c, D_c^{-1})}{8 W_B} \right\} > 0,$$

and (ii) for $c \leq (n-1)/4$ the constants are explicit, with $\kappa_c \geq e^{-c/2} \kappa_0(n)$, $\kappa_0(2) = 1$, D_c bounded and $W_B \leq 2^{1-n} \sqrt{2\pi/c} e^{(n-1)^2/(2c)}$, so that $\lambda_1(c) \geq C_n \sqrt{c} e^{-(n-1)^2/(2c)}$ for all $0 < c \leq (n-1)/4$ and some $C_n > 0$.

Part (i) is positive at every c , and part (ii), with Theorem 3.6, gives $\lambda_1(c) \asymp \sqrt{c} e^{-1/(2c)}$ on $\mathbb{H}^2$, exponent and $\sqrt{c}$ prefactor both sharp. On $\mathbb{H}^3$ the two exponents agree; for $n \geq 4$ they differ, and the fits above indicate that in these dimensions the lower bound must improve. The proof is in Appendix B.6; Table 8 (Appendix C) compares the bound with the finite element values.

Remark 3.8 (Strict positivity for every c). $\lambda_1(c) > 0$ for all $c > 0$: with $V = e^{\alpha r}$ smoothed near o , $\mathcal{L}_c V/V \rightarrow -\infty$, and the Lyapunov criterion of [19] gives a Poincaré inequality (Appendix B). The threshold changes the scale of the gap, not its positivity.

4 From the theorem to a sampler

4.1 Forward and reverse processes

The forward process follows $dX_t = -\nabla W_c dt + \sqrt{2} dB_t$ from the data law until time T , approaching μ_c at a rate set by $\lambda_1(c)$; μ_c has uniform angular component and radial density ρ_c , sampled by inverse transform. The reverse process is the classical time reversal [14, 15], on a manifold as in [3]:

$$dY_t = [\nabla W_c(Y_t) + 2\nabla \log p_{T-t}(Y_t)] dt + \sqrt{2} d\bar{B}_t, \quad (2)$$

with p_t the density with respect to vol. At stationarity $\nabla \log \mu_c = -\nabla W_c$, so the bracket is $-\nabla W_c$ and the stationary process is its own reversal. By Corollary 3.4, the last component of the data law to be forgotten by the forward process is its first angular harmonic, which the reverse process must therefore reconstruct from the least informative endpoint.

4.2 Anchored scores read in a frame

We build this stationary structure into the score parameterisation:

$$s_\theta(x, t) = -\nabla W_c(x) + u_\theta(x, t), \quad \text{and the reverse drift is } -\nabla W_c + 2u_\theta. \quad (3)$$

When $u_\theta \equiv 0$ the sampler is exactly the ergodic stationary diffusion, so training perturbs a dynamics that already confines samples, whereas an unanchored network must cancel the outward $+\nabla W_c$ term in (2) at every radius and a systematic error leaves a net outward drift. The anchor is a structural guarantee outside the training support, where the residual is unreliable: whatever the network outputs there, the confining term remains. The representation of the residual matters at large radius, where the hyperboloid embedding is poorly conditioned. For the Minkowski projection $P_x w = w + \langle x, w \rangle_L x$ onto $T_x \mathbb{H}^n$ one has $\sinh r \leq \sup_{|w|_2=1} |P_x w|_g \leq \sqrt{2} \cosh r$ (Lemma B.1). An extrinsic network output of Euclidean size $O(1)$ can therefore produce a tangent vector of size $\Theta(e^r)$, and a geodesic step inherits the same amplification. The amplification creates a feedback loop: once a sample leaves the training support, an inaccurate extrinsic output amplified by $\Theta(e^r)$ moves it to still larger radius. We therefore predict scalar coefficients in an orthonormal frame: on $\mathbb{H}^2$, away from the origin, $u_\theta = a_\theta e_r + b_\theta e_\varphi$ with $e_r = \nabla r$, whose norm is set by the network outputs and does not grow with the embedding coordinates (inputs are bounded functions of (r, φ, t) , Appendix C); for $n \geq 3$ a radial scalar head and a tangential vector head projected onto the tangent space of S^{n-1} serve the same purpose. In the one seed ablation of Table 13 (Appendix D) an untrained network sends 99%, 96% and 0% of the samples beyond $r = 6$ with the unanchored extrinsic, anchored extrinsic and frame heads. Training removes the extrinsic divergence within a few thousand iterations.

4.3 Training, discretisation and floating point

We train u_θ on simulated forward trajectories using score matching [16] in its denoising form [17], with tangent Gaussian geodesic random walks over short intervals as in Riemannian diffusion models [3]. For a transition from x_k to x_{k+m} over $\tau = mh$ we use the target $\hat{s}(x_{k+m}) = \log_{x_{k+m}}(\Phi_\tau(x_k))/(2\tau)$, with Φ_τ the drift flow, as a first order approximation to the conditional mean. With $m = 1$ the signal to noise ratio is of order $\sqrt{h}$, and the $\sinh^{-1} r$ metric factor suppresses angular displacements at large radius. We therefore draw m uniformly from $\{1, 2, 4, 8\}$, trading a target error of first order in mh from the drift flow approximation for a signal to noise gain of order $\sqrt{m}$. The wrapped baseline of Section 5.2 uses exact Gaussian conditionals (Appendix C).

Both processes are integrated by geodesic Euler–Maruyama with reprojection after each step, $x_{k+1} = \exp_{x_k}(-\nabla W_c(x_k)h + \sqrt{2h}\xi_k)$. Hyperboloid coordinates grow as e^r , so evaluating $\langle x, x \rangle_L = -1$ cancels with error $\varepsilon_{\text{mach}} e^{2r}$ and the usable radius is $r \lesssim \frac{1}{2} \log(2/\varepsilon_{\text{mach}})$, about 8.3 in fp32 and 18.4 in fp64 (Table 12, Appendix C). Our implementation therefore uses fp64, reprojects after every step and supplies the radius directly to the frame based features.

5 Experiments

5.1 Relaxation rates across the threshold

On $\mathbb{H}^2$ we integrate the forward SDE in polar coordinates, $dr = (\coth r - cr) dt + \sqrt{2} dW_r$ and $d\varphi = (\sqrt{2}/\sinh r) dW_\varphi$ (double precision, fixed seeds, $N = 20,000$ particles, $T = 6$, $h = 0.005$, $c \in \{0.05, \dots, 4\}$, small initial cloud at radius 3 and angle 0), and fit the decay rate of $R(t) = |\mathbb{E} e^{i\varphi t}|$ on the window $0.05 \leq R \leq 0.7$. By Corollary 3.4 the late time decay rate of $R(t)$ is $\lambda_1(c)$ for an initial law with a component along the first harmonic ground state. The fit therefore estimates the gap itself: the ratios to $\lambda_1(c)$ are 0.99, 0.96, 0.99 and 0.99 at $c = 0.5, 1, 2$ and 4 (Table 5). At $c = 0.25$

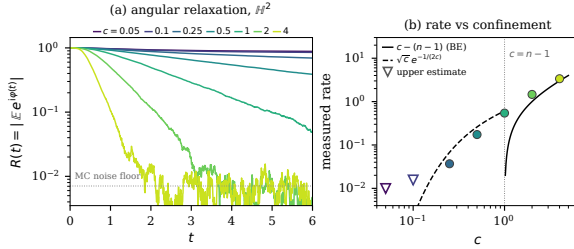


Figure 1: Forward relaxation on $\mathbb{H}^2$. (a) Angular order parameter $R(t)$ for confinements spanning the threshold $c = 1$; dotted line: the Monte Carlo floor $N^{-1/2}$. (b) Fitted rates against c : above threshold they approach the Bakry-Émery bound $c - (n-1)$ (solid) from above, with ratios 1.46 at $c = 2$ and 1.12 at $c = 4$; below threshold they collapse along the $\sqrt{c} e^{-1/(2c)}$ envelope of Theorem 3.6 (dashed). Open triangles at $c \leq 0.1$ are transient limited upper estimates.

Table 1: Hierarchy A across the confinement sweep: mean $\pm$ standard deviation over three training seeds (each the mean of three sample sets) and, for the mismatch, over three forward banks; $\lambda_1(c)T$ counts relaxation times in the horizon $T = 4$. Appendix D: hierarchies B, C, W and $\mathbb{H}^3$.

	intrinsic, confinement c					tangent
	0.25	0.5	1	2	4	(wrapped)
$\lambda_1(c)T$	0.15	0.70	2.26	5.89	13.65	–
MMD ² ($\times 10^3$)	8.8 ± 1.0	3.3 ± 0.5	2.2 ± 0.2	3.0 ± 0.6	6.6 ± 0.7	5.6 ± 0.3
leaf TV	0.133 ± 0.008	0.072 ± 0.010	0.040 ± 0.003	0.047 ± 0.007	0.063 ± 0.012	0.039 ± 0.013
mismatch ($\times 10^3$)	13.1 ± 1.2	4.5 ± 1.3	0.3 ± 0.2	0.0 ± 0.0	0.1 ± 0.1	n/a

the fitting window is entered only at $t \approx 5.8$, too late for a precise rate. Longer runs to $T = 120$ recover the rate, giving 0.039 and 0.040 against $\lambda_1 = 0.038$ (Appendix C).

Figure 1 separates the two regimes. Above the threshold the rates lie above the Bakry-Émery line and approach it as c grows; below it they fall to 0.174 at $c = 0.5$ and 0.037 at $c = 0.25$, against 0.26 and 0.068 for $\sqrt{c} e^{-1/(2c)}$ and 0.176 and 0.038 for the computed gap, and for the two smallest c the trajectories do not mix within the horizon. Observables built on the finite element sector eigenfunctions test Proposition 3.3 directly. On $\mathbb{H}^n$ for $n = 2, 3, 4$ the angular and radial ones relax at $\lambda_1(c)$ and $\lambda_1(c) + c$ to within 4%, and their rate difference divided by c lies between 0.97 and 1.02 in all 19 cases (Tables 10 and 11, Appendix C).

5.2 Generative modelling of hierarchical data

Setup. Hierarchy A is a two level hierarchy on $\mathbb{H}^2$: four groups at angles $k\pi/2$ with weights (0.4, 0.3, 0.2, 0.1), each split into two leaves at angular offsets ± 0.175 , leaf centres at radius 3.5 with wrapped Gaussian noise of scale 0.15, 8,192 training and 4,096 test points. Three further hierarchies change its geometry and scale (Appendix C): hierarchy B varies radius, depth and imbalance; hierarchy C is a tree with 64 leaves at depth three placed after [13]; hierarchy W is a synthetic mixture whose 1170 cluster centres are the WordNet mammal synsets embedded as in [12]. On each hierarchy we train the intrinsic construction at $c \in \{0.25, 0.5, 1, 2, 4\}$, spanning the threshold $c = 1$, and a tangent space baseline inspired by the hyperbolic wrapped normal [11]. In our implementation the baseline is an OU diffusion in $\log_o$ coordinates with exact denoising targets. The models share two hidden layers, Fourier features, 30k Adam iterations (40k on C and W) and 400 sampling steps, with three training seeds (five on C and W) and three sampling seeds each. We report the U-statistic estimate of MMD² with the Laplacian kernel $e^{-d/\sigma}$ on hyperbolic distance (positive definite on $\mathbb{H}^n$ [35]; Appendix C), total variation of cluster weights after nearest cluster assignment (leaf TV; node TV on W) and the fraction of diverged samples. Two forward diagnostics need no trained network: the prior mismatch, the MMD² between the forward endpoint cloud at time T and an independent sample of μ_c , and the fourth angular order parameter $R_4(t) = |\mathbb{E} e^{4i\varphi_t}|$, which measures the four fold group structure and, by Proposition 3.3(ii), relaxes no more slowly than the gap.

Effect of the threshold. With architecture, features, budget and sampler fixed, the generation error on hierarchy A is smallest at $c = 1$, the threshold itself, and rises on both sides, by a factor 4.0 at $c = 0.25$ and 3.0 at $c = 4$. The ordering $c = 1$ below $c = 2$ below $c = 4$, with $c = 0.25$ worst, holds for every training seed (Table 1, Appendix D). Below the threshold the forward diagnostics show the problem before training. At $T = 4$ the $c = 2.0$ process is mixed at the resolution of the diagnostics: the prior mismatch is zero within estimator noise and R_4 has fallen from 0.765 to 0.010, the Monte Carlo floor. At $c = 0.25$, by contrast, the radius has nearly equilibrated ($\mathbb{E}r = 3.91$ from 3.5, equilibrium 4.19), whereas the angular structure has not: $R_4(T) = 0.380$. The prior mismatch is 13.1×10^{-3} at $c = 0.25$, against 4.5 at $c = 0.5$ and 0.3 at $c = 1$, on the scale $e^{-2\lambda_1(c)T}$ with $\lambda_1(c)T = 0.15, 0.70$ and 2.3 . Above the threshold the mismatch is zero, and the rise of the error is consistent with a harder score estimation or sampling problem. The drifts, the residual to learn and the target bias of Section 4.3 grow with c , while the step cap and the discretisation do not account for the rise (Appendix D). The threshold $c = n - 1$ is therefore a crossover in the spectral gap, not a generative threshold: at a fixed horizon the error degrades once the prior mismatch exceeds the score error, which here happens below the threshold.

Scaling the hierarchy and the dimension. Hierarchy B repeats the pattern of A, and the larger hierarchies C and W, with 64 and 1170 clusters, repeat the shape of the sweep (Tables 14 and 16, Figure 3). Their MMD^2 is smallest at or just above the threshold, although on C the values at $c = 0.5, 1$ and 2 are within noise of each other. C and W also expose a resolution limit: no model at this budget, the wrapped baseline included, resolves the clusters within a branch of C or a family of W, and the diagnostics of Appendix D did not change this. On $\mathbb{H}^3$ (threshold $c = 2$) the sweep $c \in \{0.5, 1, 2, 4, 8\}$ repeats the picture, with the minimum at $c = 1$ and a threefold rise from $c = 2$ to $c = 8$ (Table 15). No sample diverged: the tangential head appears stable without a global frame.

Scope. The experiments use $n = 2$ and 3 , five hierarchies and CPU sized networks. The wrapped baseline has a larger MMD^2 than the intrinsic model at its best confinement on A and B, a similar one on C, W and $\mathbb{H}^3$ and the better leaf TV on B and $\mathbb{H}^3$. These comparisons are descriptive, since the best confinement is selected on the same test values. The baseline plausibly benefits from the bounded radii of the data, where the distortion of the $\log_o$ chart is bounded, and the comparisons therefore support feasibility, not superiority. Four narrower conclusions hold: simulated rates reproduce the spectral gap; the confinement changes generative quality on both sides of the threshold, with the minimum at or within a factor two of it; slow mixing below the threshold is visible before training; and the frame head is immune by construction to the amplification of Lemma B.1 that makes untrained extrinsic heads diverge (Table 13, Appendix D).

6 Limitations and open problems

Two main questions about the spectral gap remain open, one on each side of the threshold $c = n - 1$. Below it, the question is the exact exponent of the collapse for $n \geq 4$. The fits of Section 3.4 favour the angular exponent $2(n - 2)$ of the upper bound in Theorem 3.6. The lower bound of Theorem 3.7 misses the angular cost and has the larger exponent $(n - 1)^2/2$. The tightness of $\text{Var}_{\mu_c}(r)^{-1} - c$ for $n = 4, 5$ (Table 6) suggests that a variance expansion of the radial gap $\lambda_1(c) + c$ may reach the exact exponent. Above the threshold, the question is whether $\lambda_1(c) - c$ converges to $-\frac{2}{3}(n - 1)$ as $c \rightarrow \infty$. The formal expansion of Appendix B.9 predicts this limit and, with its $1/c$ term, is confirmed numerically to four digits. Theorem 3.5 proves this value only as an upper bound on the limit superior, and the harmonic approximation [24] may supply a proof. On the generative side, a direct link between $\lambda_1(c)$ and generative error would need control of mixing, score estimation error and the bias of the multi-span target, perhaps through a manifold Girsanov bound. The experiments are limited in scale: the WordNet based hierarchy is used at half the radius of its embedding (Appendix C), and the full WordNet noun hierarchy [12] needs angular features that grow with the number of clusters. Other geometries need their own confinement or boundary mechanism, and dynamic data, hierarchical through summation constraints, need a model over sequences of points on $\mathbb{H}^n$.

References

- [1] J. Ho, A. N. Jain, and P. Abbeel. Denoising Diffusion Probabilistic Models. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, pages 6840–6851, 2020.
- [2] Y. Song, J. Sohl-Dickstein, D. P. Kingma, A. Kumar, S. Ermon, and B. Poole. Score-Based Generative Modeling through Stochastic Differential Equations. In *International Conference on Learning Representations (ICLR)*, 2021.
- [3] V. De Bortoli, E. Mathieu, M. Hutchinson, J. Thornton, Y. W. Teh, and A. Doucet. Riemannian Score-Based Generative Modelling. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35, pages 2406–2422, 2022.
- [4] C.-W. Huang, M. Aghajohari, A. J. Bose, P. Panangaden, and A. C. Courville. Riemannian Diffusion Models. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35, pages 2750–2761, 2022.
- [5] B. Jing, G. Corso, J. Chang, R. Barzilay, and T. Jaakkola. Torsional Diffusion for Molecular Conformer Generation. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35, pages 24240–24253, 2022.
- [6] J. Yim, B. L. Trippe, V. De Bortoli, E. Mathieu, A. Doucet, R. Barzilay, and T. Jaakkola. SE(3) diffusion model with application to protein backbone generation. In *International Conference on Machine Learning (ICML)*, volume 202 of *Proceedings of Machine Learning Research*, pages 40001–40039, 2023.
- [7] J. L. Watson, D. Juergens, N. R. Bennett, et al. De novo design of protein structure and function with RFDiffusion. *Nature*, 620(7976):1089–1100, 2023.
- [8] E. Hoogeboom, V. Garcia Satorras, C. Vignac, and M. Welling. Equivariant Diffusion for Molecule Generation in 3D. In *International Conference on Machine Learning (ICML)*, volume 162 of *Proceedings of Machine Learning Research*, pages 8867–8887, 2022.
- [9] J. Austin, D. D. Johnson, J. Ho, D. Tarlow, and R. van den Berg. Structured Denoising Diffusion Models in Discrete State-Spaces. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 34, pages 17981–17993, 2021.
- [10] C. Vignac, I. Krawczuk, A. Siraudin, B. Wang, V. Cevher, and P. Frossard. DiGress: Discrete Denoising diffusion for graph generation. In *International Conference on Learning Representations (ICLR)*, 2023.
- [11] Y. Nagano, S. Yamaguchi, Y. Fujita, and M. Koyama. A Wrapped Normal Distribution on Hyperbolic Space for Gradient-Based Learning. In *International Conference on Machine Learning (ICML)*, volume 97 of *Proceedings of Machine Learning Research*, pages 4693–4702, 2019.
- [12] M. Nickel and D. Kiela. Poincaré Embeddings for Learning Hierarchical Representations. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 30, pages 6338–6347, 2017.
- [13] R. Sarkar. Low Distortion Delaunay Embedding of Trees in Hyperbolic Plane. In *Graph Drawing: 19th International Symposium, GD 2011, Eindhoven, The Netherlands, September 21–23, 2011, Revised Selected Papers*, volume 7034 of *Lecture Notes in Computer Science*, pages 355–366. Springer, 2012.
- [14] B. D. O. Anderson. Reverse-time diffusion equation models. *Stochastic Processes and their Applications*, 12(3):313–326, 1982.
- [15] U. G. Haussmann and E. Pardoux. Time Reversal of Diffusions. *Annals of Probability*, 14(4):1188–1205, 1986.
- [16] A. Hyvärinen. Estimation of Non-Normalized Statistical Models by Score Matching. *Journal of Machine Learning Research*, 6:695–709, 2005.
- [17] P. Vincent. A Connection Between Score Matching and Denoising Autoencoders. *Neural Computation*, 23(7):1661–1674, 2011.
- [18] H. P. McKean. An upper bound to the spectrum of Δ on a manifold of negative curvature. *Journal of Differential Geometry*, 4(3):359–366, 1970.

- [19] D. Bakry, F. Barthe, P. Cattiaux, and A. Guillin. A simple proof of the Poincaré inequality for a large class of probability measures including the log-concave case. *Electronic Communications in Probability*, 13:60–66, 2008.
- [20] P. A. Deift. Applications of a commutation formula. *Duke Mathematical Journal*, 45(2):267–310, 1978.
- [21] M. Reed and B. Simon. *Methods of Modern Mathematical Physics, Vol. I: Functional Analysis*. Academic Press, 1980.
- [22] M. Reed and B. Simon. *Methods of Modern Mathematical Physics, Vol. II: Fourier Analysis, Self-Adjointness*. Academic Press, 1975.
- [23] S. Karlin and H. M. Taylor. *A Second Course in Stochastic Processes*. Academic Press, 1981.
- [24] B. Simon. Semiclassical analysis of low lying eigenvalues. I. Non-degenerate minima: asymptotic expansions. *Annales de l’Institut Henri Poincaré, Section A*, 38(3):295–308, 1983. Erratum: 40(2):224, 1984.
- [25] F. Klein. *Vergleichende Betrachtungen über neuere geometrische Forschungen: Programm zum Eintritt in die philosophische Facultät und den Senat der k. Friedrich-Alexanders-Universität zu Erlangen*. Verlag von Andreas Deichert, Erlangen, 1872.
- [26] M. M. Bronstein, J. Bruna, T. Cohen, and P. Veličković. Geometric Deep Learning: Grids, Groups, Graphs, Geodesics, and Gauges. arXiv:2104.13478, 2021.
- [27] M. P. do Carmo. *Riemannian Geometry*. Birkhäuser, Boston, 1992.
- [28] I. Chavel. *Eigenvalues in Riemannian Geometry*. Academic Press, 1984.
- [29] H. Iwaniec. *Spectral Methods of Automorphic Forms*, 2nd edition. Graduate Studies in Mathematics, vol. 53. American Mathematical Society, 2002.
- [30] A. Grigor’yan. *Heat Kernel and Analysis on Manifolds*. AMS/IP Studies in Advanced Mathematics, vol. 47. American Mathematical Society and International Press, 2009.
- [31] G. Da Prato and J. Zabczyk. *Ergodicity for Infinite Dimensional Systems*. London Mathematical Society Lecture Note Series, vol. 229. Cambridge University Press, 1996.
- [32] D. Bakry, I. Gentil, and M. Ledoux. *Analysis and Geometry of Markov Diffusion Operators*. Grundlehren der mathematischen Wissenschaften, vol. 348. Springer, 2014.
- [33] T. Bröcker and T. tom Dieck. *Representations of Compact Lie Groups*. Graduate Texts in Mathematics, vol. 98. Springer, 1985.
- [34] H. J. Brascamp and E. H. Lieb. On extensions of the Brunn–Minkowski and Prékopa–Leindler theorems, including inequalities for log concave functions, and with an application to the diffusion equation. *Journal of Functional Analysis*, 22(4):366–389, 1976.
- [35] A. Feragen, F. Lauze, and S. Hauberg. Geodesic exponential kernels: When curvature and linearity conflict. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 3032–3042, 2015.

A Invariant measures across geometries: statements, proofs and tables

This appendix states and proves Theorem A.2, summarised in Section 2, and records the classification that places $\mathbb{H}^n$ among the geometries used in diffusion modelling, under the unit Laplacian convention $P_t = e^{t\Delta}$ for Brownian motion. The organising object is the symmetry group, following the viewpoint of Klein’s Erlangen programme [25, 26]. Once symmetry has identified the natural measure and spectral basis, three structural questions remain: is the volume finite, is there a boundary and is the state space continuous? Table 2 gives the short version; Tables 3 and 4 apply the classification to eleven spaces.

Proposition A.1 (Volume is stationary). *On a manifold without boundary, vol_g is infinitesimally invariant for Brownian motion. If Brownian motion is conservative, vol_g is invariant for the semigroup.*

Proof. For $f \in C_c^\infty(M)$, the divergence theorem gives $\int_M \Delta f d\text{vol}_g = \int_M \text{div}(\nabla f) d\text{vol}_g = 0$, which is infinitesimal invariance. If Brownian motion is conservative, symmetry of the heat semigroup gives $\int_M P_t f d\text{vol}_g = \int_M f P_t 1 d\text{vol}_g = \int_M f d\text{vol}_g$. $\square$

Theorem A.2 (A dichotomy between finite and infinite volume). *Let (M, g) be connected and without boundary, and assume Brownian motion is conservative. (i) If $\text{vol}_g(M) < \infty$, the normalised measure $\bar{\mu} = \text{vol}_g / \text{vol}_g(M)$ is the unique invariant probability measure of (X_t) . (ii) If, in addition, M is compact, then $-\Delta$ has discrete spectrum $0 = \lambda_0 < \lambda_1 \leq \dots$ and $\|P_t f - \bar{\mu}(f)\|_{L^2(\bar{\mu})} \leq e^{-\lambda_1 t} \|f - \bar{\mu}(f)\|_{L^2(\bar{\mu})}$: the process mixes exponentially at the spectral gap. (iii) If $\text{vol}_g(M) = \infty$, there is no invariant probability measure of constant density. Let V be smooth with $e^{-V} \in L^1(\text{vol}_g)$, and suppose the diffusion generated by $\mathcal{L}_V = \Delta - \nabla V \cdot \nabla$ is conservative. Then $d\mu_V \propto e^{-V} d\text{vol}_g$ is its unique invariant probability law.*

Proof of Theorem A.2. (i) Proposition A.1, normalisation and the assumed conservativity give invariance of $\bar{\mu}$. Brownian motion on a connected manifold is irreducible: its heat kernel $p_t(x, y)$ is strictly positive for $t > 0$ and $x, y \in M$, by the strong maximum principle for the heat operator $\partial_t - \Delta$ [30]. The heat kernel is smooth, so the semigroup is strong Feller as well as irreducible; the standard strong Feller–irreducibility criterion therefore gives uniqueness of the invariant probability measure [31].

(ii) On a compact manifold $-\Delta$ has compact resolvent, hence discrete spectrum [28] with an $L^2(\bar{\mu})$ -orthonormal eigenbasis $\{\phi_k\}_{k \geq 0}$, $-\Delta \phi_k = \lambda_k \phi_k$, $\lambda_0 = 0$ (the constants), $\lambda_1 > 0$. Writing $f = \sum_k c_k \phi_k$ with $c_0 = \int f d\bar{\mu}$, we have $P_t f = \sum_k e^{-\lambda_k t} c_k \phi_k$, and therefore $\|P_t f - c_0\|_{L^2}^2 = \sum_{k \geq 1} e^{-2\lambda_k t} |c_k|^2 \leq e^{-2\lambda_1 t} \sum_{k \geq 1} |c_k|^2$; take square roots.

(iii) A constant density has infinite total mass, which prevents vol_g from being normalised. For the drifted process, $\mathcal{L}_V$ is symmetric with respect to $d\mu_V = e^{-V} d\text{vol}_g$, since $e^V \text{div}(e^{-V} \nabla f) = \Delta f - \nabla V \cdot \nabla f = \mathcal{L}_V f$. Hence, for $f \in C_c^\infty(M)$, $\int_M \mathcal{L}_V f d\mu_V = \int_M \text{div}(e^{-V} \nabla f) d\text{vol}_g = 0$. Together with the assumed conservativity, this gives invariance; irreducibility gives uniqueness as in (i). See [32] for the general theory. On $\mathbb{R}^n$, choosing $V(x) = \frac{c}{2}|x|^2$ gives $\mu_V = \mathcal{N}(0, c^{-1}I)$, the stationary law of $dX = -cX dt + \sqrt{2} dB$. $\square$

Proposition A.3 (Drift depends on volume; the rate is a separate, spectral matter). *Let (M, g) be connected, complete and without boundary.*

- (a) *If $\text{vol}_g(M) < \infty$, whether or not M is compact, the normalised volume is the unique invariant probability measure of Brownian motion, with no drift.*
- (b) *Convergence is exponential at rate λ_1 , the bottom of the nonzero spectrum of $-\Delta$. Compactness guarantees $\lambda_1 > 0$. Finite volume alone does not give the compact spectral picture. For example, a hyperbolic surface of finite area that is not compact has essential spectrum $[1/4, \infty)$ [29]. Across compact hyperbolic surfaces, the positive gap can be arbitrarily small [28].*

Proof. Part (a) is Theorem A.2(i). A complete manifold of finite volume has $\text{vol}_g(B(x, r)) \leq e^{r^2}$ for large r and is therefore stochastically complete [30], which makes Brownian motion conservative. Part (b) is the estimate of Theorem A.2(ii) with the spectral theorem for $-\Delta$ in $L^2(\bar{\mu})$ in place of the eigenbasis, which requires only $\lambda_1 > 0$. If $\Gamma \backslash \mathbb{H}^2$ is not compact but has finite area, the essential spectrum of $-\Delta$ is $[1/4, \infty)$ and there are at most finitely many eigenvalues below it [29]. The normalised volume is therefore stationary even though the spectrum is not discrete. $\square$

Corollary A.4 (Symmetry–volume spine). *Suppose $M = G/H$ is connected and carries a G -invariant Riemannian metric.*

- (a) *vol_g is G -invariant and is the unique G -invariant measure up to positive scale; it is invariant for Brownian motion.*
- (b) *If G is compact then M is compact; the normalised G -invariant measure is the unique stationary law and no drift is needed. When $M = G$ is a compact Lie group with a bi-invariant metric, this measure is Haar measure.*

- (c) If M is non-compact with $\text{vol}_g(M) = \infty$, the G -invariant measure is not normalisable and a prior therefore requires a drift $-\nabla V$; the resulting μ_V is invariant only under $\{\gamma \in G : V \circ \gamma = V\}$, which necessarily breaks full G -invariance.

Proof. (a) G acts by isometries and therefore preserves vol_g ; transitivity makes any two G -invariant Radon measures proportional. (b) $M = G/H$ is the continuous image of the compact group G , hence compact; apply Theorem A.2(i)–(ii) and identify the bi-invariant volume on a compact group with Haar measure. (c) The G -invariant measure is vol_g up to scale, and it is infinite by hypothesis. Suppose Brownian motion had an invariant probability law. Since isometries commute with the heat semigroup, the image of this law under each $\gamma \in G$ would be another invariant probability law. Uniqueness, from the strong Feller property and irreducibility as in Theorem A.2(i), would make it G -invariant and hence a multiple of vol_g , which is impossible since vol_g has infinite mass. A prior therefore requires a drift. An isometry γ preserves $\mu_V \propto e^{-V} \text{vol}_g$ if and only if $V \circ \gamma = V$. $\square$

On $\mathbb{H}^n$, the sheet preserving isometry group $O^+(n, 1)$ preserves the infinite hyperbolic volume, whereas the drift $-c\nabla(\frac{1}{2}r^2)$ selects a base point and reduces the symmetry to the stabiliser $O(n)$. Any normalisable radial prior of this form therefore introduces a preferred centre. It is this residual $O(n)$ symmetry that produces the sector decomposition used in Section 3.3.

Proposition A.5 (Boundary flux and Neumann reflection). *Let M have boundary ∂M with outward unit normal ν . For $f \in C^\infty(\overline{M})$, $\int_M \Delta f \, d\text{vol}_g = \int_{\partial M} \partial_\nu f \, dS$. Consequently vol_g is invariant only if the boundary flux vanishes. Reflected (Neumann) Brownian motion has vol_g as an invariant measure, its generator acting on functions with $\partial_\nu f = 0$ on ∂M . A diffusion whose coefficients degenerate at ∂M may dispense with an explicit reflection. Degeneracy alone, however, decides neither whether the boundary is reached nor which measure is stationary. Both depend on the full generator, drift included, through its boundary classification.*

Proof. The identity is Green's theorem on a manifold with boundary. Stationarity for vol_g requires $\int_M \Delta f \, d\text{vol}_g = 0$ for all admissible f . By the identity this holds precisely when the boundary integral vanishes, which the Neumann condition enforces. For the degenerate case consider the Wright–Fisher (Jacobi) diffusion on $[0, 1]$ with generator $x(1-x)\partial_x^2 + (\alpha(1-x) - \beta x)\partial_x$, $\alpha, \beta > 0$. Its scale and speed densities are $x^{-\alpha}(1-x)^{-\beta}$ and $x^{\alpha-1}(1-x)^{\beta-1}$. With reflection at an endpoint that is reached, its stationary law is therefore the Beta(α, β) law, uniform only for $\alpha = \beta = 1$. By Feller's boundary classification [23], the endpoint 0 is reached with positive probability when $\alpha < 1$ and is not reached when $\alpha \geq 1$. The same holds on the simplex with the Dirichlet($\alpha_1, \dots, \alpha_n$) law in place of the Beta law. There the face on which the coordinates in a set I vanish is reached exactly when $\sum_{i \in I} \alpha_i < 1$. Degeneracy of the diffusion coefficient is therefore compatible with either accessible or inaccessible boundaries, and the stationary law depends on the drift and the boundary condition; the uniform law occurs only for $\alpha = \beta = 1$. $\square$

Remark A.6 (No manifold, no SDE). When the state space is a finite or countable set S , the generator Δ is replaced by a Markov rate matrix Q and stationarity becomes $\pi Q = 0$ for a non-explosive chain. In discrete time, with a transition matrix P , stationarity becomes $\pi P = \pi$. A group Γ acting on S is a symmetry when Q commutes with its permutation representation. Curvature, volume and boundary have no direct analogue; the relevant distinction is smooth versus discrete, with equivariance imposed on the model rather than realised by a metric.

Proposition A.7 (Spectral expansion and Peter–Weyl). *On a compact manifold, $-\Delta$ has discrete spectrum with an orthonormal eigenbasis $\{\phi_k\}$, and the heat kernel is $p_t(x, y) = \sum_k e^{-\lambda_k t} \phi_k(x) \overline{\phi_k(y)}$. Suppose further that $M = G/H$ is a compact homogeneous space whose metric is invariant under G . Then Δ commutes with the G action. Peter–Weyl theory [33] decomposes $L^2(G/H)$ into irreducible G representations that contain vectors fixed by H , and Δ acts blockwise on this decomposition. Diagonalising within the finite multiplicity blocks gives a representation theoretic eigenbasis. This recovers spherical harmonics on S^n and Fourier characters on the torus, where the decomposition is multiplicity free. On $\text{SO}(3)$ it gives the Wigner D functions, and there the representation of dimension $2\ell + 1$ occurs $2\ell + 1$ times.*

Proof. The spectral expansion is standard on compact manifolds. Since G acts by isometries, Δ commutes with the induced unitary representation and therefore preserves each isotypic component of $L^2(G/H)$. Peter–Weyl identifies the components that occur with the irreducible representations admitting vectors fixed by H . The self adjoint operator $-\Delta$ can then be diagonalised on the corresponding finite multiplicity spaces, yielding the stated eigenbasis and the familiar bases in the listed examples. $\square$

Remark A.8 (Curvature feeds compactness: Bonnet–Myers). If (M, g) is complete and $\text{Ric} \geq (n-1)\kappa g$ with $\kappa > 0$, then M is compact with diameter at most $\pi/\sqrt{\kappa}$ [27]. A strictly positive Ricci lower bound therefore forces the finite volume regime with no drift, whereas the non-compact examples $\mathbb{R}^n$ and $\mathbb{H}^n$ cannot satisfy such a bound. On $\mathbb{H}^n$, curvature sets the confinement strength needed by the Bakry–Émery estimate, which is the role studied in the main text.

Table 2: The classification in compact form. Symmetry supplies the prior and the basis; finite volume decides the drift; a boundary decides reflection; discreteness leaves the SDE regime. The $\mathbb{H}^n$ row is the subject of this paper.

Space	vol < ∞	Stationary prior	Forward process	Kernel basis
$\mathbb{R}^n$	no	Gaussian (via drift)	OU/VP SDE	Hermite; Fourier
$S^n, \text{SO}(3), T^n$	yes	uniform/Haar	Brownian motion	harmonics; Wigner D ; Fourier
$\text{SE}(3)$	no	Haar $\times$ Gaussian	BM $\times$ OU, factorised	Wigner $D \times$ Gaussian
simplex Δ^{n-1}	yes	Dirichlet (uniform for unit parameters)	Wright–Fisher, degenerate at the boundary	Jacobi polynomials
$\mathbb{H}^n$	no	$\mu_c \propto e^{-cr^2/2} d\text{vol}$ (§3)	drifted BM $\mathcal{L}_c$ (§3)	radial $\times$ spherical harmonics
SPD cone	no	drift dependent	affine invariant BM + drift	zonal functions
discrete	n/a	uniform/absorbing	Markov chain, discrete or continuous time	matrix powers or exponential
BHV tree space	no	open	open	n/a

Tables 3 and 4 apply the same classification to eleven spaces in a common row order. Table 3 records the geometry. Table 4 records the corresponding diffusion choices. The stationary measure entries of the homogeneous spaces follow Corollary A.4. The obstruction column uses Theorem A.2, Proposition A.5 and Remark A.6. The spectral column follows Proposition A.7. The $\mathbb{H}^n$ row is established in the main text.

Table 3: Geometric profile. What each space is: curvature, compactness, boundary, topology and the symmetry group that organises the rest. Row order matches Table 4.

Space	Curvature	Compact?	Boundary	Topology	Symmetry group
Euclidean space $\mathbb{R}^n$	Flat ($K = 0$)	No	None	Contractible; π_1 trivial	$E(n) = O(n) \ltimes \mathbb{R}^n$ (flat homogeneous)
n -sphere $S^n \subset \mathbb{R}^{n+1}$	Positive ($K = +1$)	Yes	None	Simply connected ($n \geq 2$); $\pi_1(S^1) = \mathbb{Z}$	$O(n+1)$; $S^n \cong O(n+1)/O(n)$
Flat torus $T^n = \mathbb{R}^n/\mathbb{Z}^n$	Flat ($K = 0$)	Yes	None	$\pi_1 = \mathbb{Z}^n$ (not simply connected)	Abelian T^n (translations)
Rotations $SO(3)$ (compact Lie group)	Non-negative (bi-invariant)	Yes	None	$\pi_1 = \mathbb{Z}_2$; $SO(3) \cong \mathbb{RP}^3$ (cover $SU(2)$)	Left/right translation; bi-invariant metric
Rigid motions $SE(3) = SO(3) \ltimes \mathbb{R}^3$	Non-negative (product metric); no bi-invariant metric	No	None	$\pi_1 = \mathbb{Z}_2$; $\mathbb{R}^3$ contractible	Left-invariant; semidirect product
Hyperbolic space $\mathbb{H}^n$	Negative ($K = -1$)	No	None (∂_∞ not a manifold boundary)	Contractible (Hadamard $\cong \mathbb{R}^n$)	$O^+(n, 1)$; $\mathbb{H}^n \cong SO^+(n, 1)/SO(n)$
SPD cone $\text{Sym}^+(n)$ (affine-invariant metric)	Non-positive (Hadamard/CAT(0))	No	None intrinsically (singular PSD at ∞)	Contractible (open convex cone)	$GL(n)$ congruence $P \mapsto APA^\top$; $\cong GL(n)/O(n)$
Probability simplex Δ^{n-1}	Positive (Fisher–Rao)	Yes	Yes: faces where some $x_i = 0$	Contractible (convex); faces form boundary	S_n permutations; $\sqrt{\cdot}$ -isometry to S^{n-1} orthant
Stiefel $V_k(\mathbb{R}^n)$; Grassmann $Gr(k, n)$	Non-negative (normal metric; Grassmann symmetric)	Yes	None	Non-trivial (e.g. $\pi_1(Gr) = \mathbb{Z}_2$, real)	$O(n)$ acts transitively; homogeneous
Graphs/categorical discrete data	N/A (not a smooth manifold)	N/A (discrete)	N/A	Combinatorial; no differential structure	S_n (node permutations/automorphisms)
Stratified/BHV tree space $\mathcal{T}_n$	CAT(0), singular	No	No manifold boundary; singular strata (collapsed edges)	Cone-complex; not a manifold	Label permutations; local orthant charts

Table 4: Diffusion prescription. What geometry forces on the process: admissible forward dynamics, the stationary prior, the obstruction and its remedy, the natural kernel basis and representative methods. Row order matches Table 3.

Space	Forward process	Stationary prior	Obstruction & remedy	Kernel/eigenbasis	Representative methods
Euclidean space $\mathbb{R}^n$	VP/OU SDE (mean-reverting); VE (variance-exploding) has no stationary law	Isotropic Gaussian $\mathcal{N}(0, I)$	Infinite volume $\rightarrow$ drift needed for a normalisable prior	Gaussian heat kernel; Hermite/Fourier	DDPM; Score-SDE
n -sphere S^n	Intrinsic heat flow/Riemannian BM (Laplace–Beltrami)	Uniform (surface measure)	None (compact); no drift required	Spherical harmonics $Y_{\ell m}$	Riemannian SGM; directional data
Flat torus T^n	Wrapped BM (heat kernel = wrapped Gaussian)	Uniform on T^n	None (compact); must respect periodicity	Fourier modes $e^{2\pi i k \cdot x}$, $k \in \mathbb{Z}^n$	Torsional diffusion (conformers)
Rotations $\text{SO}(3)$	Group BM = IGSO(3) (its heat kernel)	Haar measure (= uniform)	None (compact group): no drift	Wigner- D functions $D_{\ell m n}$	FrameDiff; RFdiffusion
Rigid motions $\text{SE}(3)$	Factorised: BM on $\text{SO}(3) \times \text{OU}$ on $\mathbb{R}^3$	$\text{Haar}(\text{SO}(3)) \times \mathcal{N}(\mathbb{R}^3)$	Infinite volume of translations $\rightarrow$ drift; rotation needs none	Wigner- $D \times$ Gaussian	SE(3) diffusion (poses, docking, robotics)
Hyperbolic space $\mathbb{H}^n$	Confined BM $\mathcal{L}_c$; other Riemannian drifts possible	$\mu_c \propto e^{-cr^2}/2 d\text{vol}$ for this confinement	Infinite volume $\rightarrow$ no Brownian stationary probability; $c > n - 1$ gives the Bakry–Émery rate	Radial functions $\times$ spherical harmonics; heat kernel (McKean for $n = 2$)	This paper; Riemannian diffusion; wrapped normal
SPD cone $\text{Sym}^+(n)$	Affine-invariant BM; log-Euclidean/Wishart-type	Log-normal/Wishart-type (via drift)	Infinite volume $\rightarrow$ drift; iterates must stay PD	Zonal/matrix spherical functions	Covariance; DTI tensors; connectivity
Probability simplex Δ^{n-1}	Wright–Fisher/Jacobi; or $\sqrt{\cdot}$ -map to sphere; or reflected Euclidean BM	Dirichlet: uniform for Wright–Fisher with unit parameters or reflected Euclidean BM, Dirichlet($\frac{1}{2}, \dots, \frac{1}{2}$) for the $\sqrt{\cdot}$ -map	Boundary $\rightarrow$ reflection or a face-preserving process	Jacobi polynomials; harmonics via $\sqrt{\cdot}$ -map	Compositional data; Dirichlet diffusion
Stiefel/Grassmann	Riemannian BM on the homogeneous space	Uniform ($O(n)$ -invariant measure)	None (compact); retract to enforce constraint	Zonal/representation-theoretic functions	Frames, subspaces, PCA-type constraints
Graphs/discrete	Markov chain noising, discrete or continuous time (uniform/absorbing/marginal): not an SDE	Uniform, marginal or δ at absorbing [MASK]	Discrete $\rightarrow$ Markov jump; impose permutation equivariance	Transition-matrix powers or exponential	D3PM; DiGress
Stratified/tree space	No canonical BM across strata (candidates: sticky/reflected)	Open: no agreed invariant law	Singular strata $\rightarrow$ manifold diffusion undefined	n/a	Frontier/open problem

B Proofs for Section 3

B.1 Proof of Theorem 3.1

Proof of Theorem 3.1. (i) In geodesic polar coordinates about o , $d\text{vol} = |S^{n-1}| \sinh^{n-1}(r) dr d\sigma$ with σ normalised. Since $\sinh^{n-1}(r) \leq Ce^{(n-1)r}$ whereas $e^{-cr^2/2}$ has Gaussian decay, μ_c has finite mass. The generator is symmetric in $L^2(\mu_c)$ and can be written as $\mathcal{L}_c = e^{W_c} \text{div}(e^{-W_c} \nabla \cdot)$. It follows that $\int \mathcal{L}_c f d\mu_c = 0$ for $f \in C_c^\infty(\mathbb{H}^n)$ by the divergence theorem. The semigroup is symmetric in $L^2(\mu_c)$ and conservative, as shown at the end of this proof. Hence $\int P_t f d\mu_c = \int f P_t 1 d\mu_c = \int f d\mu_c$, which proves invariance of μ_c . The heat kernel of this uniformly elliptic diffusion is smooth and strictly positive on the connected manifold, so the semigroup is strong Feller and irreducible; uniqueness follows from the same criterion [31]. Finally, r^2 is a Lyapunov function for the confining drift and rules out explosion. (ii) On $\mathbb{H}^n$, $\text{Ric} = -(n-1)g$, while the Hessian comparison theorem on a Cartan–Hadamard manifold gives $\nabla^2(\frac{1}{2}r^2) \geq g$ [27]. Hence $\nabla^2 W_c \geq cg$ and $\text{Ric} + \nabla^2 W_c \geq (c - (n-1))g$, after which the Bakry–Émery criterion yields the Poincaré inequality with constant $c - (n-1)$ and the stated decay [32]. $\square$

Throughout, $d\mu_c = Z_c^{-1} \sinh^{n-1}(s) e^{-cs^2/2} ds d\sigma(\theta)$ in geodesic polar coordinates, with σ the normalised measure on S^{n-1} and $Z_c = \int_0^\infty \sinh^{n-1}(s) e^{-cs^2/2} ds$. We use repeatedly that for $s \geq 1$,

$$\gamma e^s \leq \sinh s \leq \frac{1}{2}e^s, \quad \gamma := \frac{1}{2}(1 - e^{-2}), \quad (4)$$

and that $\sinh x \leq x e^{x^2/6}$ for all $x \geq 0$ (compare the series termwise: $(2k+1)! \geq 6^k k!$).

B.2 Proof of Theorem 3.2

The Rayleigh bound. The function $f = r$ is 1-Lipschitz with $|\nabla r| = 1$ away from o , lies in $L^2(\mu_c)$ and belongs to the form domain of $\mathcal{L}_c$ (Lipschitz functions do; see [32]). The variational characterisation gives

$$\lambda_1(c) \leq \frac{\int |\nabla r|^2 d\mu_c}{\text{Var}_{\mu_c}(r)} = \frac{1}{\text{Var}_{\mu_c}(r)}.$$

The radial variance at strong confinement. Under μ_c the radius has density $\rho_c(s) = Z_c^{-1} \sinh^{n-1}(s) e^{-cs^2/2}$ on $(0, \infty)$. Let $U = \sqrt{c}r$, with density proportional to

$$h_c(u) = c^{(n-1)/2} \sinh^{n-1}(u/\sqrt{c}) e^{-u^2/2}.$$

Pointwise $h_c(u) \rightarrow u^{n-1} e^{-u^2/2}$ as $c \rightarrow \infty$, and by $\sinh x \leq x e^{x^2/6}$,

$$h_c(u) \leq u^{n-1} e^{(n-1)u^2/(6c)} e^{-u^2/2} \leq u^{n-1} e^{-u^2/3} \quad (c \geq n-1),$$

an integrable dominating function that remains integrable against u and u^2 . Dominated convergence applied to the normalisation and to the first two moments gives $\mathbb{E}U \rightarrow \mathbb{E}\chi_n$ and $\mathbb{E}U^2 \rightarrow \mathbb{E}\chi_n^2 = n$, where χ_n has density proportional to $u^{n-1} e^{-u^2/2}$ and $\mathbb{E}\chi_n = \sqrt{2} \Gamma(\frac{n+1}{2}) / \Gamma(\frac{n}{2})$. Hence

$$c \text{Var}_{\mu_c}(r) = \text{Var}(U) \longrightarrow v_n = n - 2 \left(\frac{\Gamma((n+1)/2)}{\Gamma(n/2)} \right)^2.$$

Positivity of v_n holds because χ_n is not deterministic. For the upper bound $v_n \leq 1$, write $\chi_n = |Z|$ with $Z \sim \mathcal{N}(0, I_n)$. The map $z \mapsto |z|$ is 1-Lipschitz, and the Gaussian Poincaré inequality gives $\text{Var}(|Z|) \leq \mathbb{E}|\nabla|z||^2 = 1$.

Conclusion. Choose $c_0(n)$ such that $\text{Var}_{\mu_c}(r) \geq v_n/(2c)$ for $c \geq c_0(n)$. Then $\lambda_1(c) \leq 2c/v_n$, which with Theorem 3.1(ii) sandwiches $\lambda_1(c)$ between two linear functions of c . $\square$

B.3 Proof of Proposition 3.3

Sector decomposition. The rotations of $\mathbb{H}^n$ about o act by isometries that fix r , so they commute with $\mathcal{L}_c$ and preserve μ_c . Expanding $f(s, \theta) = \sum_\ell \sum_j g_{\ell j}(s) Y_{\ell j}(\theta)$ in an orthonormal basis of spherical harmonics, with $-\Delta_{S^{n-1}} Y_{\ell j} = \ell(\ell + n - 2) Y_{\ell j}$, the Dirichlet form separates:

$$\int |\nabla f|^2 d\mu_c = \sum_{\ell, j} \int_0^\infty \left(g_{\ell j}^2 + \frac{\ell(\ell + n - 2)}{\sinh^2 s} g_{\ell j}^2 \right) \rho_c ds = \sum_{\ell, j} \mathcal{E}_\ell(g_{\ell j}),$$

since $|\nabla f|^2 = (\partial_s f)^2 + \sinh^{-2}(s) |\nabla_{S^{n-1}} f|^2$ in polar coordinates. Hence $-\mathcal{L}_c$ is the orthogonal sum of the operators $-\mathcal{L}_c^{(\ell)}$ of (1), each acting on $\mathcal{H} = L^2((0, \infty), \rho_c ds)$ and repeated with the multiplicity $\dim \mathcal{Y}_\ell$ of the harmonics of degree ℓ . Part (ii) follows immediately, because $\mathcal{E}_\ell \geq \mathcal{E}_1$ on a common form domain for every $\ell \geq 1$.

Factorisation. Let $D = \partial_s$ on $\mathcal{H}$ with domain $\{F \in \mathcal{H} : F \text{ locally absolutely continuous, } F' \in \mathcal{H}\}$, which is the radial part of the form domain of $\mathcal{L}_c$. Then D is closed and densely defined, $\|DF\|^2 = \mathcal{E}_0(F)$, and the self adjoint operator associated with this closed form is $D^*D = -\mathcal{L}_c^{(0)}$. For $g \in C_c^\infty((0, \infty))$, integrating by parts against the weight ρ_c and using $\rho'_c = a\rho_c$ gives

$$\langle DF, g \rangle_{\mathcal{H}} = \int_0^\infty F' g \rho_c ds = - \int_0^\infty F (g' + ag) \rho_c ds, \quad \text{so} \quad D^*g = -g' - ag.$$

For $g \in \text{dom}(D^*)$, a second integration by parts on $[\varepsilon, \infty)$ gives

$$\|D^*g\|^2 = \lim_{\varepsilon \rightarrow 0} \int_\varepsilon^\infty (g'^2 + a^2 g^2 + a(g^2)') \rho_c ds = \lim_{\varepsilon \rightarrow 0} \left(\int_\varepsilon^\infty (g'^2 - a'g^2) \rho_c ds + [a g^2 \rho_c]_\varepsilon^\infty \right),$$

because $(a\rho_c)' = (a' + a^2)\rho_c$. Since $-a'(s) = (n-1)\sinh^{-2}s + c$, the integrand is $g'^2 + (n-1)\sinh^{-2}(s)g^2 + cg^2$. The boundary term at infinity vanishes along a sequence $s_k \uparrow \infty$, because $|a(s)| \leq Cs$ for large s and $\int_0^\infty g^2 \rho_c ds < \infty$. The boundary term at the origin satisfies $a\rho_c = \rho'_c \sim (n-1)Z_c^{-1}s^{n-2}$ as $s \rightarrow 0$, and it vanishes along a sequence $\varepsilon_k \downarrow 0$ for every g with $\int_0^\infty g^2 \sinh^{-2}(s) \rho_c ds < \infty$, which is the finiteness of $\mathcal{E}_1(g)$. Letting $\varepsilon \rightarrow 0$ along this sequence,

$$\|D^*g\|^2 = \mathcal{E}_1(g) + c\|g\|^2, \quad \text{that is} \quad DD^* = -\mathcal{L}_c^{(1)} + c,$$

where the identification of $\text{dom}(D^*)$ with the form domain of $\mathcal{E}_1$ is justified in the next paragraph.

Domains. Three facts make the identification of the self adjoint realisations precise. First, $\mathcal{E}_0(F) = \|DF\|^2$ with D closed. Hence $\mathcal{E}_0$ is a closed form on $\{F \in \mathcal{H} : F' \in \mathcal{H}\}$, and its operator is D^*D by the representation theorem for closed forms [21]. The space $C_c^\infty((0, \infty))$ is a core for $\mathcal{E}_0$ because the origin is a polar set for $n \geq 2$. Polarity of the origin allows a function in the form domain to be cut off near the origin at a cost in $\|F'\|$ that tends to zero, and the Gaussian weight makes cutoffs at infinity harmless. Second, DD^* is self adjoint for every closed densely defined D [22], and the computation above shows that it acts as $-\mathcal{L}_c^{(1)} + c$ on $C_c^\infty((0, \infty))$. Third, $-\mathcal{L}_c^{(1)}$ is essentially self adjoint on $C_c^\infty((0, \infty))$ in $\mathcal{H}$, by Weyl's limit point criterion at both endpoints [22]. At $s = 0$ the equation $-\mathcal{L}_c^{(1)}g = \lambda g$ has indicial roots 1 and $1 - n$. The solution behaving like s^{1-n} is not square integrable against $\rho_c \sim Z_c^{-1}s^{n-1}$ for $n \geq 2$. At $s = \infty$ the ground state transform $g \mapsto \rho_c^{1/2}g$ carries $-\mathcal{L}_c^{(1)}$ to the Schrödinger operator $-\partial_s^2 + a^2/4 + a'/2 + (n-1)\sinh^{-2}s$ of Appendix B.9. Its potential is bounded below and grows like $c^2s^2/4$. Hence DD^* is the unique self adjoint realisation of $-\mathcal{L}_c^{(1)} + c$ and coincides with the operator of the closed form $\mathcal{E}_1$, whose form domain is $\text{dom}(D^*)$. The same transform shows that the potential tends to $+\infty$ at both ends, like $(n^2 - 1)/(4s^2)$ at the origin. Hence $-\mathcal{L}_c^{(1)}$ has compact resolvent and discrete spectrum. Its bottom $\lambda^{(1)}(c)$ is a simple eigenvalue with a positive eigenfunction, the ground state of the degree one sector used in Corollary 3.4. For the radial sector with $n \in \{2, 3\}$ the origin is limit circle. In this case D^*D is the realisation selected by the form domain above, which contains the constants and corresponds to the solution bounded at the origin. This realisation is the radial part of $-\mathcal{L}_c$ acting on smooth functions on $\mathbb{H}^n$.

Isospectrality. For a closed densely defined operator A between Hilbert spaces, the polar decomposition $A = U|A|$ makes A^*A restricted to $(\ker A)^\perp$ unitarily equivalent to AA^* restricted to $(\ker A^*)^\perp$; in particular the nonzero spectra of A^*A and AA^* coincide with multiplicity [21]. Here $\ker D$ consists of the constants, which carry the eigenvalue 0 of $-\mathcal{L}_c^{(0)}$, and $\ker D^* = \{g : (g\rho_c)' = 0\} = \{C/\rho_c\} \cap \mathcal{H} = \{0\}$ because $\int_0^\infty \rho_c^{-2} \rho_c ds = \infty$. Therefore the nonzero spectrum of $-\mathcal{L}_c^{(0)}$ equals the spectrum of $-\mathcal{L}_c^{(1)} + c$, which is part (i). The eigenfunction statement is the same identity at the level of solutions: differentiating $F'' + aF' = -\Lambda F$ gives $(F')'' + a(F')' + a'F' = -\Lambda F'$, that is $-\mathcal{L}_c^{(1)} F' = (\Lambda - c)F'$ by the expression for $-a'$. Part (iii) combines (i), (ii) and $c > 0$: the bottom of the spectrum of $-\mathcal{L}_c$ on the orthogonal complement of the constants is $\min\{\lambda^{(1)} + c, \lambda^{(1)}, \lambda^{(2)}, \dots\} = \lambda^{(1)}$. $\square$

On $\mathbb{R}^n$ the same computation with $\rho_c(s) \propto s^{n-1}e^{-cs^2/2}$ gives $-a' = (n-1)s^{-2} + c$ and reproduces the OU ladder, in which the degree one sector has bottom c and the radial sector has bottom $2c$. On $\mathbb{H}^n$ the map $F \mapsto F'\theta_1$ survives, but the degree one bottom is no longer c ; Theorem 3.6 shows that it is exponentially small below the threshold. Since $-(\log \rho_c)'' = -a' \geq c$, the radial density is log-concave with constant c , so the one dimensional Bakry-Émery argument gives $\lambda^{(0)}(c) \geq c$ for every $c > 0$, in agreement with $\lambda^{(0)}(c) = c + \lambda_1(c)$ and $\lambda_1(c) > 0$.

Proof of Corollary 3.4. The test function r lies in the radial sector, so the Rayleigh bound of Theorem 3.2 applies to $\lambda^{(0)}(c)$; subtracting c by Proposition 3.3(iii) gives the first inequality. The lim inf follows from Theorem 3.1(ii) and the lim sup from $c\text{Var}_{\mu_c}(r) \rightarrow v_n$. The decay statements follow from the sector decomposition and from (iii). $\square$

B.4 Proof of Theorem 3.5

The test function. In the hyperboloid model $x = (\sinh r \theta, \cosh r)$ with $\theta \in S^{n-1}$, let $x_1 = \sinh(r)\theta_1$, where θ_1 is a linear coordinate restricted to the sphere. The reflection $\theta_1 \mapsto -\theta_1$ is an isometry fixing o , so it preserves μ_c and $\mu_c(x_1) = 0$. In geodesic polar coordinates $|\nabla f|^2 = (\partial_r f)^2 + \sinh^{-2}(r)|\nabla_S f|^2$, and $|\nabla_S \theta_1|^2 = 1 - \theta_1^2$ on the sphere, so

$$|\nabla x_1|^2 = \cosh^2(r)\theta_1^2 + 1 - \theta_1^2 = 1 + \sinh^2(r)\theta_1^2.$$

Both $x_1^2 \leq \sinh^2 r$ and $|\nabla x_1|^2 \leq 1 + \sinh^2 r$ are integrable against $e^{-cr^2/2} \sinh^{n-1} r$, so x_1 lies in the form domain. Under μ_c the angle is uniform on S^{n-1} and independent of r , with $\mathbb{E}\theta_1^2 = 1/n$. Hence

$$\int |\nabla x_1|^2 d\mu_c = 1 + \frac{\mathbb{E}_{\mu_c}[\sinh^2 r]}{n}, \quad \text{Var}_{\mu_c}(x_1) = \frac{\mathbb{E}_{\mu_c}[\sinh^2 r]}{n},$$

and the variational characterisation gives $\lambda_1(c) \leq 1 + n/\mathbb{E}_{\mu_c}[\sinh^2 r]$.

Scaling. As in the proof of Theorem 3.2, let $U = \sqrt{c}r$ with density proportional to $h_c(u) = c^{(n-1)/2} \sinh^{n-1}(u/\sqrt{c})e^{-u^2/2}$, and let $h_\infty(u) = u^{n-1}e^{-u^2/2}$. Write $\varphi(x) = \sinh(x)/x \geq 1$, so that $h_c = \varphi(u/\sqrt{c})^{n-1}h_\infty$ and $c \sinh^2(u/\sqrt{c}) = u^2\varphi(u/\sqrt{c})^2$. Then $\mathbb{E}_{\mu_c}[\sinh^2 r] = S(c)/c$ with

$$S(c) := \frac{\int_0^\infty u^2 \varphi(u/\sqrt{c})^2 h_c(u) du}{\int_0^\infty h_c(u) du},$$

so the bound equals $1 + nc/S(c)$. Since $nc/S(c) = c(S(c)/n)^{-1}$ and $1 - \frac{2n+1}{3} = -\frac{2(n-1)}{3}$, the claim $1 + nc/S(c) = c - \frac{2(n-1)}{3} + o(1)$ is equivalent to

$$S(c) = n + \frac{n(2n+1)}{3c} + o(c^{-1}) \quad (c \rightarrow \infty). \quad (5)$$

Two dominated limits. From $\sinh x \leq xe^{x^2/6}$ we have $1 \leq \varphi(x) \leq e^{x^2/6}$, and $e^a - 1 \leq ae^a$ for $a \geq 0$ gives, for every integer $k \geq 1$ and $u \geq 0$,

$$0 \leq c(\varphi(u/\sqrt{c})^k - 1) \leq \frac{k}{6} u^2 e^{ku^2/(6c)}, \quad \text{and} \quad c(\varphi(u/\sqrt{c})^k - 1) \rightarrow \frac{k}{6} u^2,$$

the limit from $\varphi(x) = 1 + x^2/6 + O(x^4)$. For $c \geq \frac{2(n+1)}{3}$ the factors φ^{n-1} and φ^2 together contribute at most $e^{(n+1)u^2/(6c)} \leq e^{u^2/4}$, so every integrand below is dominated by a constant multiple of $(1 + u^{n+5})e^{-u^2/4}$. Write $m_k = \int u^k h_\infty du$ and $m_k(c) = \int u^k h_c du$. Dominated convergence gives, for $k \in \{0, 2\}$,

$$c(m_k(c) - m_k) = \int_0^\infty u^k c(\varphi(u/\sqrt{c})^{n-1} - 1) h_\infty(u) du \rightarrow \frac{n-1}{6} m_{k+2},$$

and, writing the numerator of $S(c)$ as $m_2(c) + \int u^2(\varphi(u/\sqrt{c})^2 - 1)h_c du$,

$$c \int_0^\infty u^2(\varphi(u/\sqrt{c})^2 - 1) h_c(u) du \rightarrow \frac{1}{3} m_4.$$

Conclusion. The moments of h_∞ are those of χ_n , so $m_2/m_0 = n$ and $m_4/m_0 = n(n+2)$. Hence

$$\begin{aligned} c\left(\frac{m_2(c)}{m_0(c)} - n\right) &= \frac{c(m_2(c) - m_2)m_0 - m_2 c(m_0(c) - m_0)}{m_0(c)m_0} \\ &\rightarrow \frac{n-1}{6} \left(\frac{m_4}{m_0} - n \frac{m_2}{m_0}\right) = \frac{n(n-1)}{3}, \end{aligned}$$

while the second limit gives $c(S(c) - m_2(c)/m_0(c)) \rightarrow \frac{1}{3} m_4/m_0 = \frac{n(n+2)}{3}$. Adding the two, $c(S(c) - n) \rightarrow \frac{n(n-1)}{3} + \frac{n(n+2)}{3} = \frac{n(2n+1)}{3}$, which is (5). The two sided statement follows from Theorem 3.1(ii). $\square$

B.5 Proof of Theorem 3.6

Test function. Let $\psi(\theta) = \sqrt{n} \theta_1$ on S^{n-1} (for $n = 2$, $\psi(\varphi) = \sqrt{2} \cos \varphi$): a first spherical harmonic with

$$\int \psi d\sigma = 0, \quad \int \psi^2 d\sigma = 1, \quad \int |\nabla_{S^{n-1}} \psi|^2 d\sigma = n-1,$$

the last since ψ is an eigenfunction of the spherical Laplacian with eigenvalue $n-1$. Let η be a fixed smooth cutoff with $\eta = 0$ on $[0, 1]$, $\eta = 1$ on $[2, \infty)$, $0 \leq \eta \leq 1$, $|\eta'| \leq 2$, and set $f(x) = \eta(r)\psi(\theta)$: smooth and bounded with bounded gradient, since $|\nabla f|^2 = \eta'(r)^2 \psi^2 + \eta(r)^2 |\nabla_S \psi|^2 / \sinh^2 r$ and η vanishes near the origin.

Mean and variance. $\int f d\mu_c = (Z_c^{-1} \int \eta \sinh^{n-1} e^{-cs^2/2} ds) (\int \psi d\sigma) = 0$, and

$$\text{Var}_{\mu_c}(f) = Z_c^{-1} \int_1^\infty \eta^2 \sinh^{n-1}(s) e^{-cs^2/2} ds \geq \mu_c^{\text{rad}}([2, \infty)).$$

We claim $q_n := \inf_{0 < c \leq 1} \mu_c^{\text{rad}}([2, \infty)) > 0$. Indeed $c \mapsto \mu_c^{\text{rad}}([2, \infty))$ is continuous and strictly positive on $(0, 1]$ (dominated convergence in both numerator and Z_c), while as $c \downarrow 0$,

$$\mu_c^{\text{rad}}([0, 2]) \leq Z_c^{-1} \int_0^2 \sinh^{n-1}(s) ds \leq Z_c^{-1} e^{2(n-1)} \rightarrow 0$$

by the lower bound (6) on Z_c below, and hence the function tends to 1 at the open end. A positive continuous function on $(0, 1]$ with positive limit inferior at 0 has positive infimum.

Energy. Using the two identities for ψ ,

$$\begin{aligned} \int |\nabla f|^2 d\mu_c &= Z_c^{-1} \left[\int_1^2 \eta'^2 \sinh^{n-1} e^{-cs^2/2} ds + (n-1) \int_1^\infty \eta^2 \sinh^{n-3}(s) e^{-cs^2/2} ds \right] \\ &\leq Z_c^{-1} \left[4e^{2(n-1)} + (n-1)A_c \right], \end{aligned}$$

where $A_c := \int_1^\infty \sinh^{n-3}(s) e^{-cs^2/2} ds$ and we bounded $\int_1^2 \sinh^{n-1} \leq \sinh^{n-1}(2) \leq e^{2(n-1)}$.

The partition function from below. By (4), for $0 < c \leq 1$,

$$Z_c \geq \gamma^{n-1} \int_1^\infty e^{(n-1)s - cs^2/2} ds = \gamma^{n-1} e^{\frac{(n-1)^2}{2c}} \int_1^\infty e^{-\frac{c}{2}(s - \frac{n-1}{c})^2} ds \geq \frac{1}{2} \gamma^{n-1} \sqrt{\frac{2\pi}{c}} e^{\frac{(n-1)^2}{2c}}, \quad (6)$$

since the Gaussian centre $(n-1)/c \geq n-1 \geq 1$ lies in the integration range, the integral contains at least half the Gaussian mass.

The angular integral from above. For $n = 2$, (4) gives $\sinh^{-1}(s) \leq \gamma^{-1}e^{-s}$ on $[1, \infty)$, hence $A_c \leq \gamma^{-1}e^{-1}$, a constant. For $n \geq 3$, $\sinh^{n-3}(s) \leq 2^{-(n-3)}e^{(n-3)s}$ and completing the square,

$$A_c \leq 2^{-(n-3)} \sqrt{\frac{2\pi}{c}} e^{\frac{(n-3)^2}{2c}}.$$

Conclusion. Combining, $\lambda_1(c) \leq q_n^{-1} Z_c^{-1} [4e^{2(n-1)} + (n-1)A_c]$. For $n = 2$, both terms in the bracket are constants and (6) gives

$$\lambda_1(c) \leq C \sqrt{c} e^{-\frac{1}{2c}}.$$

For $n \geq 3$, the cutoff term contributes $C_1(n) \sqrt{c} e^{-(n-1)^2/(2c)} \leq C_1(n) e^{-(n-1)^2/(2c)}$, while the angular term contributes, after the $\sqrt{2\pi/c}$ factors cancel,

$$C_2(n) e^{-\frac{(n-1)^2 - (n-3)^2}{2c}} = C_2(n) e^{-\frac{2(n-2)}{c}},$$

whence $\lambda_1(c) \leq C(n) e^{-\beta(n)/c}$ with $\beta(n) = \min\{(n-1)^2/2, 2(n-2)\}$. $\square$

The proof makes the two costs of Section 3.4 literal. The first exponent is the relative mass of a unit ball at the origin against a shell at radius $(n-1)/c$. The second is the angular kinetic energy of the first harmonic, $(n-1)Z_c^{-1}A_c$, which that shell does not carry. The integrand of A_c peaks near $s = (n-3)/c$ (near $s = 1$ for $n = 3$), where the weight $\sinh^{-2}s \approx 4e^{-2s}$ is larger. The energy is of order $e^{-[(n-1)^2 - (n-3)^2]/(2c)} = e^{-2(n-2)/c}$. For $n = 3$ the two exponents coincide at $2/c$. Since $\frac{(n-1)^2}{2} - 2(n-2) = \frac{(n-3)^2}{2} \geq 0$, the angular cost $2(n-2)$ is no larger for every $n \geq 3$ and determines $\beta(n)$; it is strictly smaller once $n \geq 4$.

B.6 Proof of Theorem 3.7

Reduction to the first harmonic. By Proposition 3.3(iii), $\lambda_1(c) = \lambda^{(1)}(c)$ is the bottom of the spectrum of $-\mathcal{L}_c^{(1)}$ in (1), that is

$$\lambda_1(c) = \inf_{\eta} \frac{\int_0^\infty (\eta'^2 + (n-1)\eta^2/\sinh^2 s) w ds}{\int_0^\infty \eta^2 w ds}, \quad w = Z_c \rho_c = \sinh^{n-1}(s) e^{-cs^2/2}.$$

The normalisation Z_c cancels, and the infimum runs over the form domain of $\mathcal{E}_1$, without a mean zero constraint. It suffices to take $\eta \in C_c^\infty((0, \infty))$, a core for this form (Appendix B.3).

(i) The first harmonic. Let $V = (n-1)/\sinh^2 s$, $A = [0, 1]$, $B = [1, \infty)$ and $V_1 = (n-1)/\sinh^2 1 = \min_A V$. For such η set

$$\begin{aligned} \alpha &= \eta(1), & m &= \frac{1}{W_B} \int_B \eta w, & X &= \int_A \eta^2 w, & Y &= \int_B (\eta - m)^2 w, \\ G &= \int_B \eta'^2 w, & E_A &= \int_A (\eta'^2 + V \eta^2) w. \end{aligned}$$

The energy is $E \geq E_A + G$. The norm is $N = X + Y + m^2 W_B$ because $\int_B (\eta - m)w = 0$. Three inequalities control these quantities.

(C) By homogeneity $E_A \geq \kappa_c \alpha^2$, and $E_A \geq V_1 X$ since $V \geq V_1$ on A .

(P) The probability measure $w ds/W_B$ on B has $-(\log w)'' = c + (n-1)/\sinh^2 s \geq c$, so it is log-concave with constant c on the interval B . The weights $w e^{-k(1-s)^2_+}$ on $(0, \infty)$ are log-concave

with constant c and converge to this restriction to the convex set B as $k \rightarrow \infty$. The Brascamp–Lieb inequality [34] (see also [32, Theorem 4.9.1]), applied to these weights, gives $Y \leq c^{-1}G$ in the limit.

(H) Put $T(t) = \int_t^\infty w$. By Fubini, $m - \alpha = W_B^{-1} \int_B (\eta(s) - \eta(1)) w(s) ds = W_B^{-1} \int_1^\infty \eta'(t) T(t) dt$, and the Cauchy–Schwarz inequality gives $(m - \alpha)^2 \leq D_c G$. The constant is finite. The function $g = -(\log w)' = cs - (n - 1) \coth s$, which is $-a(s)$ in (1), is increasing, and where it is positive $T(s) = \int_s^\infty (-w')/g \leq w(s)/g(s)$. Hence, for $s_2 \geq 1$ with $g(s_2) > 0$, $T^2/w \leq w/g(s_2)^2$ is integrable on $[s_2, \infty)$, while on $[1, s_2]$ one has $T \leq W_B$ and w is bounded below.

Now distinguish two cases. If $|\alpha| \geq |m|/2$, then (C) and (P) give

$$E \geq \frac{1}{2}(\kappa_c \alpha^2 + V_1 X) + cY \geq \frac{1}{8}\kappa_c m^2 + \frac{1}{2}V_1 X + cY \geq \min\left\{\frac{\kappa_c}{8W_B}, \frac{V_1}{2}, c\right\} N.$$

In the other case $|m - \alpha| > |m|/2$, and (C), (H) and (P) give

$$E \geq V_1 X + \frac{1}{2}\left(\frac{(m - \alpha)^2}{D_c} + cY\right) \geq V_1 X + \frac{m^2}{8D_c} + \frac{c}{2}Y \geq \min\left\{\frac{1}{8D_c W_B}, V_1, \frac{c}{2}\right\} N.$$

Both cases give $\lambda^{(1)}(c) \geq \min\{c/2, V_1/2, \min(\kappa_c, D_c^{-1})/(8W_B)\}$, which is the bound in (i). It is positive: W_B and D_c are finite, and $\kappa_c \geq e^{-c/2}\kappa_0(n)$ with $\kappa_0(n) > 0$ by the next paragraph.

(ii) The constants. On A , $w \geq e^{-c/2} \sinh^{n-1} s$, so $\kappa_c \geq e^{-c/2}\kappa_0(n)$ with $\kappa_0(n)$ the infimum at $c = 0$. This infimum is positive: on $[\frac{1}{2}, 1]$ one has $w \geq \sinh^{n-1}(\frac{1}{2})$ and $\eta(1)^2 \leq \int_{1/2}^1 (\eta'^2 + 4\eta^2) ds$, so $\kappa_0(n) \geq \sinh^{n-1}(\frac{1}{2}) \min\{1, V_1/4\}$. For $n = 2$ the Euler–Lagrange equation $(\sinh s \eta')' = \eta/\sinh s$ has the solution $\eta = \tanh(s/2)$, regular at the origin. The functional is a positive quadratic form on an affine space, so this critical point is the minimiser. Integrating by parts gives $\kappa_0(2) = \sinh(1) \eta'(1)/\eta(1) = 1$. For D_c split the integral at $s_1 = (n - 1)/c + 1$. On $[1, s_1]$ use $T \leq W_B$. For $c \leq (n - 1)/4$ one has $cs^2/2 - (n - 1)s \leq -3(n - 1)s/8$ on this interval and $\sinh s \geq \frac{1}{2}(1 - e^{-2})e^s$, hence $\int_1^{s_1} w^{-1} ds \leq \bar{D}_n := \frac{2^{n+2}}{3(n-1)}(1 - e^{-2})^{1-n}e^{-3(n-1)/8}$. On $[s_1, \infty)$ use $T^2/w \leq w/g^2$ with $g(s) \geq g(s_1) = c - 2(n - 1)/(e^{2s_1} - 1) \geq c/2$. The last inequality holds because $4x \leq e^{2x}$ for $x = (n - 1)/c \geq 4$, and the result is $W_B^{-2} \int_{s_1}^\infty T^2/w \leq 4c^{-2}W_B^{-1}$. Finally $W_B \leq \int_0^\infty w \leq 2^{1-n} \int_{\mathbb{R}} e^{(n-1)s - cs^2/2} ds = 2^{1-n} \sqrt{2\pi/c} e^{(n-1)^2/(2c)}$ by $\sinh s \leq e^s/2$. In the other direction, the argument for (6) applies to $W_B = \int_1^\infty w$ itself for $c \leq (n - 1)/4$, since the Gaussian centre $(n - 1)/c \geq 4$ lies in $[1, \infty)$. Hence $W_B \geq \frac{1}{2}\gamma^{n-1} \sqrt{2\pi/c} e^{(n-1)^2/(2c)}$, and $4c^{-2}W_B^{-1}$ is bounded on $(0, (n - 1)/4]$. With these bounds each of the three terms in (i) is at least a constant multiple of $\sqrt{c} e^{-(n-1)^2/(2c)}$ on $(0, (n - 1)/4]$, which is (ii). $\square$

The proof splits the radial line at $s = 1$. The mass W_B of the outer region against the capacity κ_c of the unit ball is the price of the first harmonic. This price is the radial cost $(n - 1)^2/(2c)$ in the exponent of Theorem 3.6. The angular cost $2(n - 2)/c$ of that theorem does not appear, because (P) and (H) do not see the angular potential V on B . This is why the two exponents agree for $n \leq 3$ only.

B.7 The amplification lemma of Section 4.2

Lemma B.1 (Extrinsic outputs are exponentially amplified). *Let $x \in \mathbb{H}^n$ with $r = d(o, x)$ and for $w \in \mathbb{R}^{n+1}$ let $P_x w = w + \langle x, w \rangle_L x$ be the Minkowski projection onto $T_x \mathbb{H}^n$. Then*

$$\sinh r \leq \sup_{|w|_2=1} |P_x w|_g \leq \sqrt{2} \cosh r,$$

the lower value attained at the timelike unit vector. An extrinsic network output of Euclidean size $O(1)$ can therefore produce a tangent vector of size $\Theta(e^r)$. A geodesic step inherits the same amplification.

Proof of Lemma B.1. Using $\langle x, x \rangle_L = -1$, $|P_x w|_g^2 = \langle w, w \rangle_L + \langle x, w \rangle_L^2$. For $w = e_{n+1}$ this equals $-1 + \cosh^2 r = \sinh^2 r$, while in general $\langle w, w \rangle_L \leq |w|_2^2 = 1$. Cauchy–Schwarz, applied to w with its last coordinate negated, gives $\langle x, w \rangle_L^2 \leq |x|_2^2 |w|_2^2 = \cosh 2r$; hence $|P_x w|_g^2 \leq 1 + \cosh 2r = 2 \cosh^2 r$. $\square$

B.8 Positivity for every $c > 0$ (Remark 3.8)

Fix $\alpha > 0$ and let $V = \chi(r)e^{\alpha r}$ with χ a smoothing that makes V of class C^2 near o and $V \geq 1$. Away from o , using $\Delta r = (n-1) \coth r$ and $|\nabla r| = 1$,

$$\frac{\mathcal{L}_c V}{V} = \alpha^2 + \alpha(n-1) \coth r - \alpha c r \xrightarrow{r \rightarrow \infty} -\infty.$$

There are therefore $\theta > 0$, $b < \infty$ and $R < \infty$ with $\mathcal{L}_c V \leq -\theta V + b \mathbf{1}_{B_R}$. This Lyapunov condition, together with the local Poincaré inequality on balls, implies a global Poincaré inequality for μ_c by the criterion of [19]; hence $\lambda_1(c) > 0$ for every $c > 0$.

B.9 The large c expansion

Remark B.2 (Above the threshold: the Euclidean limit). For large c the law μ_c concentrates at scale $c^{-1/2}$ about o , where $\coth s = s^{-1} + s/3 + O(s^3)$ and $\sinh^{-2} s = s^{-2} - \frac{1}{3} + O(s^2)$. Inserting these expansions in (1) with $\ell = 1$ gives

$$-\mathcal{L}_c^{(1)} = \left[-\partial_s^2 - \left(\frac{n-1}{s} - \left(c - \frac{n-1}{3} \right) s \right) \partial_s + \frac{n-1}{s^2} \right] - \frac{n-1}{3} + \mathcal{E}_{\text{rem}},$$

where the bracket is the degree one sector of the Euclidean OU operator with confinement $c - (n-1)/3$, whose bottom eigenvalue is $c - (n-1)/3$ with eigenfunction s , and where $\mathcal{E}_{\text{rem}}$ collects the higher order terms, which are of relative size $O(1/c)$ at scale $c^{-1/2}$. Carrying the expansion to second order (derivation below) gives

$$\lambda_1(c) = c - \frac{2}{3}(n-1) + \frac{4(n-1)(n+2)}{45c} + O(c^{-2}), \quad \lambda^{(0)}(c) = \lambda_1(c) + c,$$

the second by Proposition 3.3(iii). The computation is short because s is an exact eigenfunction of the first order perturbation, so the second order coefficient is a Gaussian moment ratio. The expansion is formal: a rigorous version would follow the harmonic approximation for low lying eigenvalues [24], which we do not carry out, so we state it as a prediction. Theorem 3.5 (Appendix B.4) proves the leading term as an upper bound, $\limsup_{c \rightarrow \infty} (\lambda_1(c) - c) \leq -\frac{2}{3}(n-1)$, and the $1/c$ term remains a prediction. Numerically it holds. On $\mathbb{H}^n$ for $n = 2, \dots, 5$, the finite element values of $\lambda_1(c) - c$ at $c = 256$ are $-0.6653, -1.3299, -1.9938$ and -2.6569 against the predicted $-\frac{2}{3}(n-1) = -0.6667, -1.3333, -2$ and -2.6667 , and the scaled remainder $c(\lambda_1(c) - c + \frac{2}{3}(n-1))$ converges as $c \rightarrow \infty$ to $0.356, 0.889, 1.600$ and 2.489 , against $4(n-1)(n+2)/45 = 0.356, 0.889, 1.600$ and 2.489 . In particular $\lambda_1(c)/c \rightarrow 1$, the Bakry–Émery bound $c - (n-1)$ of Theorem 3.1(ii) is sharp to leading order with a deficit tending to $(n-1)/3$, and the upper end $v_n^{-1} - 1$ of the ratio interval in Corollary 3.4 is not attained.

Derivation. Substitute $s = y/\sqrt{c}$ in (1) with $\ell = 1$ and expand $\coth(y/\sqrt{c}) = \sqrt{c}/y + y/(3\sqrt{c}) - y^3/(45c^{3/2}) + O(c^{-5/2})$ and $\sinh^{-2}(y/\sqrt{c}) = c/y^2 - \frac{1}{3} + y^2/(15c) + O(c^{-2})$, with $\partial_s = \sqrt{c} \partial_y$. Collecting powers of c ,

$$-\mathcal{L}_c^{(1)} = cT_0 + T_1 + c^{-1}T_2 + O(c^{-2}),$$

$$T_0 = -\partial_y^2 - \left(\frac{n-1}{y} - y \right) \partial_y + \frac{n-1}{y^2}, \quad T_1 = -\frac{n-1}{3} (y \partial_y + 1), \quad T_2 = (n-1) \left(\frac{y^3}{45} \partial_y + \frac{y^2}{15} \right).$$

T_0 is the degree one sector of the Euclidean OU operator with unit confinement. It is symmetric on $L^2(w_0 dy)$ with $w_0(y) = y^{n-1} e^{-y^2/2}$, its lowest eigenvalue 1 is simple, and the eigenfunction is $g_0(y) = y$. Write $\lambda_1(c) = c\lambda^{[0]} + \lambda^{[1]} + c^{-1}\lambda^{[2]} + O(c^{-2})$ and expand the eigenfunction accordingly. Since T_1 and T_2 are not symmetric on $L^2(w_0 dy)$, the Rayleigh–Schrödinger coefficients are computed with the left eigenvector of T_0 in $L^2(dy)$: symmetry of T_0 on $L^2(w_0 dy)$ means that its $L^2(dy)$ adjoint is $w_0 T_0 w_0^{-1}$, so the left eigenvector for the eigenvalue 1 is $w_0 g_0$, and pairing with it is the weighted inner product, $\int_0^\infty w_0 g_0 (T_k g) dy = \langle g_0, T_k g \rangle_{w_0}$ with $\langle f, g \rangle_{w_0} = \int_0^\infty f g w_0 dy$. The coefficients are therefore $\lambda^{[1]} = \langle g_0, T_1 g_0 \rangle_{w_0} / \langle g_0, g_0 \rangle_{w_0}$ and $\lambda^{[2]} = (\langle g_0, T_2 g_0 \rangle_{w_0} + \langle g_0, (T_1 - \lambda^{[1]}) g_1 \rangle_{w_0}) / \langle g_0, g_0 \rangle_{w_0}$, where g_1 solves $(T_0 - 1)g_1 = -(T_1 - \lambda^{[1]})g_0$ with $\langle g_0, g_1 \rangle_{w_0} = 0$; the

symmetric case identities that move T_1 to the other side of an inner product are not available, but they are not needed below. First, $\lambda^{[0]} = 1$. Second, $T_1 g_0 = -\frac{n-1}{3}(y+y) = -\frac{2}{3}(n-1)g_0$ exactly, so $\lambda^{[1]} = -\frac{2}{3}(n-1)$ and the first order correction to the eigenfunction vanishes, because the equation $(T_0 - 1)g_1 = -(T_1 - \lambda^{[1]})g_0 = 0$ has only the solution $g_1 \in \text{span}\{g_0\}$. Third, with $g_1 = 0$ the second order coefficient reduces to

$$\lambda^{[2]} = \frac{\langle g_0, T_2 g_0 \rangle_{w_0}}{\langle g_0, g_0 \rangle_{w_0}} = \frac{4(n-1)}{45} \frac{\int_0^\infty y^{n+3} e^{-y^2/2} dy}{\int_0^\infty y^{n+1} e^{-y^2/2} dy} = \frac{4(n-1)(n+2)}{45},$$

using $T_2 g_0 = (n-1)(\frac{1}{45} + \frac{1}{15})y^3 = \frac{4(n-1)}{45}y^3$ and $\int_0^\infty y^m e^{-y^2/2} dy = 2^{(m-1)/2} \Gamma(\frac{m+1}{2})$, so that the moment ratio equals $2\Gamma(\frac{n+4}{2})/\Gamma(\frac{n+2}{2}) = n+2$. The radial statement follows from Proposition 3.3(iii); directly, the radial sector has $T_1^{(0)} = -\frac{n-1}{3}y\partial_y$ acting on the radial ground state $y^2 - n$, which is not an eigenfunction of $T_1^{(0)}$, and the intertwining identity is the shorter route.

Cross-check by the ground state transform. The same coefficients follow from a symmetric problem in which no left eigenvector is needed. The unitary map $f \mapsto \rho_c^{1/2} f$ from $L^2(\rho_c ds)$ to $L^2(ds)$ carries $-\mathcal{L}_c^{(1)}$ to the Schrödinger operator $H = -\partial_s^2 + a^2/4 + a'/2 + (n-1)\sinh^{-2}s$. In the variable y , $H = cH_0 + H_1 + c^{-1}H_2 + O(c^{-2})$ with

$$H_0 = -\partial_y^2 + \frac{y^2}{4} - \frac{n}{2} + \frac{n^2 - 1}{4y^2}, \quad H_1 = -\frac{n-1}{6}(y^2 - (n-2)), \quad H_2 = (n-1)\left(\frac{y^4}{90} + \frac{(n+1)y^2}{60}\right),$$

all symmetric on $L^2(dy)$. H_0 is the radial harmonic oscillator with eigenvalues $1 + 2k$ and ground state $u_0 = y^{(n+1)/2}e^{-y^2/4}$, under which y^2 and y^4 have means $n+2$ and $(n+2)(n+4)$. Here H_1 does not have u_0 as an eigenfunction, so the first order correction to the eigenfunction is nonzero and the second order coefficient carries a sum over states. Since $(y^2 - (n+2))u_0$ is proportional to the first excited state u_1 , that sum has a single term with energy denominator 2, and

$$\lambda^{[1]} = -\frac{(n-1)(n+2)}{6} + \frac{(n-1)(n-2)}{6} = -\frac{2}{3}(n-1),$$

$$\lambda^{[2]} = \frac{(n-1)(n+2)(n+4)}{90} + \frac{(n-1)(n+1)(n+2)}{60} - \frac{(n-1)^2(n+2)}{36} = \frac{4(n-1)(n+2)}{45},$$

using $\text{Var}_{u_0}(y^2) = 2(n+2)$ in the last term. The two routes distribute the contributions differently and agree; the supplied script `check_rs.py` carries out the second route symbolically.

The expansion is formal in two respects: the series for $\coth$ and $\sinh^{-2}$ converge only for $|y| < \pi\sqrt{c}$, and T_1, T_2 are unbounded perturbations of T_0 . Both are standard features of the harmonic approximation for low lying eigenvalues, for which [24] gives the asymptotic expansion in the Schrödinger setting, but we have not carried the argument through for the present operator.

C Experimental details

The core experiment script uses NumPy in double precision on a single CPU, with Matplotlib for plotting and SciPy, SymPy and mpmath in the auxiliary verification scripts. The reported experiment scripts run in under an hour at the stated CPU scale. Seeds are fixed in the supplementary code.

C.1 Relaxation rates (Section 5.1) and numerical spectral gaps (Table 5)

Integrator. We integrate the forward SDE on $\mathbb{H}^2$ in exact polar coordinates: $dr = (\coth r - cr)dt + \sqrt{2}dW_r$ and $d\varphi = (\sqrt{2}/\sinh r)dW_\varphi$. Negative Euler proposals for r are reflected at $r = 0$, a numerical treatment of the Bessel-type singularity. We use $N = 20,000$ particles, $h = 0.005$ and $T = 6$, clipping the drift at ± 50 , which binds only near $r = 0$. Initially, $r \sim |\mathcal{N}(3, 0.2^2)|$ and $\varphi \sim \mathcal{N}(0, (0.2/\sinh 3)^2)$, which gives a small cloud at radius 3 and angle 0.

Table 5: Spectral gap of $-\mathcal{L}_c$ on $\mathbb{H}^2$ against the bounds of Section 3. Numerical values are the lowest eigenvalues of the sector problems (1) by a linear finite element method (grid refinement changes them by less than 10^{-5} relative). Simulated rates are from Figure 1; the one at $c = 0.25$ is transient limited (see the text). The mark n/a denotes an unreached fitting window ($c = 0.1$), a confinement not simulated ($c = 8$), a vacuous bound or a confinement outside the range $c \leq 1$ of Theorem 3.6.

c	0.1	0.25	0.5	1	2	4	8
$\lambda^{(1)}(c) = \lambda_1(c)$, numerical	0.00127	0.0379	0.176	0.565	1.474	3.412	7.375
$\lambda^{(0)}(c) - c$, numerical	0.00127	0.0379	0.176	0.565	1.474	3.412	7.375
simulated rate (Section 5.1)	n/a	0.0369	0.174	0.544	1.464	3.368	n/a
lower bound $c - (n - 1)$	n/a	n/a	n/a	0	1	3	7
upper bound $\text{Var}_{\mu_c}(r)^{-1} - c$ (Cor. 3.4)	0.00146	0.0491	0.239	0.775	2.011	4.612	9.899
envelope $\sqrt{c}e^{-1/(2c)}$ (Thm. 3.6)	0.00213	0.0677	0.260	0.607	n/a	n/a	n/a

Observable and fits. We estimate $R(t) = |\mathbb{E}e^{i\varphi_t}|$ over particles and take rates as least squares slopes of $\log R$ on the common window $0.05 \leq R \leq 0.7$, with $t > 0.05$; the same window is used for every c . For $c \in \{0.05, 0.1\}$ the trajectory does not enter this window within the horizon, and the plotted open triangles therefore use slopes over $[T/2, T]$; they are only transient limited upper estimates, as stated in the caption. The small c envelope $\sqrt{c}e^{-1/(2c)}$ explains why the window is not reached: it predicts a time constant of order 5×10^2 at $c = 0.1$, two orders beyond the simulated horizon. The confinement $c = 0.25$ is a borderline case. The trajectory enters the window only at $t \approx 5.8$ ($R(6) = 0.695$), and the rate in Table 5 therefore rests on the last 36 points over 0.18 time units. The whole horizon is only about a quarter of one relaxation time $1/\lambda_1 = 26$. This rate is consistent with λ_1 , although it is not a measurement at the precision of the other entries; the replicate started at the radial equilibrium gives 0.044. Longer runs settle the cases $c = 0.1$ and $c = 0.25$. With the same integrator, initial law, N and h (`expl_long.py`, two replicates each, slopes on the same window), they give 0.0386 and 0.0397 at $c = 0.25$ and $T = 120$ (ratios 1.02 and 1.05 to $\lambda_1 = 0.0379$; about 680 points over 68 time units). At $c = 0.5$ and $T = 40$ they give 0.178 and 0.180 (ratios 1.01 and 1.02), and at $c = 0.1$ and $T = 2400$, where the window is entered at $t \approx 116$, they give 0.00121 and 0.00127 (ratios 0.95 and 1.00).

Cross-checks. Two further checks test the initial law and the integrator. A replicate initialised at the radial equilibrium $r_*(c)$, the root of $\coth r = cr$, and fitted on $t \geq 1$ with $R > 0.035$, reproduces the fitted rates within about 20% at $c = 0.25, 0.5, 1$ and 2 (the largest deviation, 0.044 against 0.037, is at $c = 0.25$). At the other confinements that window gives no fit. An independent integrator in full hyperboloid coordinates, geodesic Euler–Maruyama with projection after every step, agrees with the polar one at $c = 1$. With 4,000 particles from the same initial law, it gives $R(T) = 0.053$ against 0.049 from the polar integrator, a difference within the Monte Carlo standard error of about 0.01 at that sample size.

Numerical spectral gaps. The values in Table 5 are the lowest eigenvalues of the sector forms $\mathcal{E}_\ell$ of (1), obtained by a linear finite element discretisation on a uniform grid over $[0, R]$ with $R = \max\{9, 1/c + 9/\sqrt{c}\}$ and 8,000 elements, with two point Gauss quadrature per element and the weight ρ_c evaluated in log space. The radial sector uses natural boundary conditions and the second eigenvalue is reported; the angular sectors use a Dirichlet condition at $r = 0$ and the first eigenvalue is reported. Doubling the number of elements changes the values by less than 10^{-5} relative. The radial variance $\text{Var}_{\mu_c}(r)$ in the bound of Corollary 3.4 is computed by quadrature of ρ_c on the same interval. The exponent fits quoted in Section 3.4 use the same discretisation with 12,000 elements and $R = \max\{9, (n-1)/c + 10/\sqrt{c}\}$; they are least squares slopes of $\log \lambda_1(c)$ against $1/c$ over $c \in \{0.05, \dots, 0.125\}$ for $n = 2$ (with $\frac{1}{2} \log c$ subtracted), $c \in \{0.10, \dots, 0.22\}$ for $n = 3$, $c \in \{0.18, \dots, 0.35\}$ for $n = 4$ and $c \in \{0.26, \dots, 0.45\}$ for $n = 5$. The verification scripts require NumPy 2.

Table 6: Spectral gap of $-\mathcal{L}_c$ on $\mathbb{H}^n$ for $n = 3, 4, 5$ from the sector problems (1), with the bounds of Section 3. The residual row tests Proposition 3.3(i). Entries marked n/a are values of c at which a bound is vacuous or the expansion is negative. The two entries below 10^{-5} are solved in extended precision.

c	0.25	0.5	1	2	4	8	16
$\mathbb{H}^3$ ($\beta(3) = 2$, threshold $c = 2$)							
$\lambda_1(c)$, numerical	7.91×10^{-4}	0.0346	0.268	1.032	2.869	6.773	14.72
residual $\lambda^{(0)} - \lambda^{(1)} - c$	-4×10^{-11}	5×10^{-10}	1×10^{-8}	6×10^{-8}	4×10^{-7}	2×10^{-6}	8×10^{-6}
upper bound $\text{Var}_{\mu_c}(r)^{-1} - c$	8.38×10^{-4}	0.0403	0.330	1.282	3.532	8.263	17.86
ratio $\lambda_1/(\text{Var}^{-1} - c)$	0.943	0.859	0.812	0.806	0.812	0.820	0.824
lower bound $c - (n - 1)$	n/a	n/a	n/a	0	2	6	14
large c expansion $c - \frac{2}{3}(n - 1)$	n/a	n/a	n/a	0.667	2.667	6.667	14.67
$\mathbb{H}^4$ ($\beta(4) = 4$, threshold $c = 3$)							
$\lambda_1(c)$, numerical	1.23×10^{-6}	2.74×10^{-3}	0.101	0.681	2.375	6.195	14.10
residual $\lambda^{(0)} - \lambda^{(1)} - c$	-4×10^{-14}	-5×10^{-11}	2×10^{-9}	3×10^{-8}	2×10^{-7}	1×10^{-6}	5×10^{-6}
upper bound $\text{Var}_{\mu_c}(r)^{-1} - c$	1.23×10^{-6}	2.86×10^{-3}	0.114	0.800	2.786	7.205	16.30
ratio $\lambda_1/(\text{Var}^{-1} - c)$	0.996	0.959	0.882	0.851	0.853	0.860	0.865
lower bound $c - (n - 1)$	n/a	n/a	n/a	n/a	1	5	13
large c expansion $c - \frac{2}{3}(n - 1)$	n/a	n/a	n/a	0	2	6	14
$\mathbb{H}^5$ ($\beta(5) = 6$, threshold $c = 4$)							
$\lambda_1(c)$, numerical	6.04×10^{-10}	9.38×10^{-5}	0.0284	0.419	1.933	5.642	13.49
residual $\lambda^{(0)} - \lambda^{(1)} - c$	-5×10^{-19}	6×10^{-12}	1×10^{-10}	1×10^{-8}	1×10^{-7}	8×10^{-7}	4×10^{-6}
upper bound $\text{Var}_{\mu_c}(r)^{-1} - c$	6.04×10^{-10}	9.43×10^{-5}	0.0302	0.472	2.199	6.378	15.15
ratio $\lambda_1/(\text{Var}^{-1} - c)$	1.000	0.995	0.942	0.887	0.879	0.885	0.890
lower bound $c - (n - 1)$	n/a	n/a	n/a	n/a	0	4	12
large c expansion $c - \frac{2}{3}(n - 1)$	n/a	n/a	n/a	n/a	1.333	5.333	13.33

Higher dimensions. Table 6 repeats the computation on $\mathbb{H}^3$, $\mathbb{H}^4$ and $\mathbb{H}^5$ with 12,000 elements and $R = \max\{9, (n - 1)/c + 10/\sqrt{c}\}$. The residual $\lambda^{(0)} - \lambda^{(1)} - c$ of Proposition 3.3 is below 10^{-5} in absolute value in every entry (below 1.5×10^{-4} on $\mathbb{H}^2$ at $c = 16$), and it grows with c because the eigenfunctions sharpen at scale $c^{-1/2}$ on a fixed grid. Doubling the number of elements changes the entries by less than 10^{-7} relative. The two entries below 10^{-5} are the same discretisation solved in 40 digit arithmetic (`check_extended.py`), since rounding dominates there in double precision. On $\mathbb{H}^5$ at $c = 0.25$ the double precision solve returns 6.01×10^{-10} and moves by up to 2% under refinement. The extended solve gives 6.035×10^{-10} , unchanged to eight digits at 24,000 elements and 0.99996 of the bound. The large c values quoted in Appendix B.9 use $R = \min\{9, 1 + 12/\sqrt{c}\}$ with 16,000 elements, and the limits of the scaled remainder are Richardson extrapolations from $c = 128$ and $c = 256$; doubling the number of elements changes the eigenvalues by less than 3×10^{-9} .

Checks of Theorems 3.5 and 3.7. Tables 7, 8 and 9 use the same discretisation with 16,000 elements (24,000 for Table 9) and $R = \max\{(n - 1)/c + 12/\sqrt{c}, 2\}$ (`fem_lambda.py`). Doubling the number of elements changes the eigenvalues above 10^{-6} by less than 10^{-6} relative, whereas smaller ones are limited by rounding in double precision. The three below 10^{-8} , on $\mathbb{H}^3$ at $c = 0.1$ in Table 8 and at $c = 0.025$ and 0.03 in Table 9, are therefore the same discretisation solved in 40 digit arithmetic (`check_extended.py`). Table 7 tests Theorem 3.5: it compares $\lambda_1(c)$ with the lower bound $c - (n - 1)$ of Theorem 3.1 and the upper bound $1 + n/\mathbb{E}_{\mu_c}[\sinh^2 r]$, the expectation computed by quadrature (`verify3.py`, which also checks the identity $\Delta x_1 = n x_1$ symbolically). The difference $\lambda_1(c) - c$ approaches the constant $-2(n - 1)/3$ from above and is within 0.0014 of it on $\mathbb{H}^2$ and 0.0035 on $\mathbb{H}^3$ at $c = 256$. This convergence is the formal prediction of Appendix B.9; Theorem 3.5 proves only the upper bound. Table 8 tests Theorem 3.7, listing its constants with the resulting bound and the gap (`verify4.py`, which also checks the inequality (H) of Appendix B.6 on random functions). The capacity κ_c comes from the Euler–Lagrange equation of its functional, checked against a finite element minimisation, and D_c and W_B come from quadrature. On $\mathbb{H}^2$ the ratio $\lambda_1(c)/\text{bound}$ stays between 5.6 and 7.0 from $c = 1$ down to $c = 0.05$, which shows that the bound carries both the exponent and the $\sqrt{c}$ prefactor. On $\mathbb{H}^3$ the ratio grows from 10 at $c = 1$ to

Table 7: Strong confinement: $\lambda_1(c)$ from the sector problems (1) against the lower bound $c - (n - 1)$ of Theorem 3.1 and the upper bound $1 + n/\mathbb{E}_{\mu_c}[\sinh^2 r]$ of Theorem 3.5. The difference $\lambda_1(c) - c$ approaches the value $-2(n - 1)/3$ predicted by the formal expansion of Appendix B.9. Theorem 3.5 proves this value as an upper bound on the limit superior.

c	$\mathbb{H}^2, -2(n - 1)/3 = -0.667$				$\mathbb{H}^3, -2(n - 1)/3 = -1.333$			
	$c - (n - 1)$	$\lambda_1(c)$	bound	$\lambda_1(c) - c$	$c - (n - 1)$	$\lambda_1(c)$	bound	$\lambda_1(c) - c$
2	1	1.4736	1.7466	-0.5264	0	1.0324	1.4413	-0.9676
4	3	3.4118	3.5395	-0.5882	2	2.8691	3.0517	-1.1309
8	7	7.3750	7.4352	-0.6250	6	6.7729	6.8544	-1.2271
32	31	31.3443	31.3585	-0.6557	30	30.6942	30.7123	-1.3058
128	127	127.3361	127.3396	-0.6639	126	126.6736	126.6780	-1.3264
256	255	255.3347	255.3365	-0.6653	254	254.6701	254.6723	-1.3299

Table 8: The constants of Theorem 3.7, its bound and the finite element value of $\lambda_1(c)$; the last column is the ratio $\lambda_1(c)/\text{bound}$.

n	c	κ_c	D_c	W_B	bound of Theorem 3.7	$\lambda_1(c)$	ratio
2	0.05	0.9882	0.8865	1.23×10^5	1.00×10^{-6}	7.01×10^{-6}	7.01
2	0.1	0.9766	1.0308	586.8	2.07×10^{-4}	1.27×10^{-3}	6.16
2	0.25	0.9425	1.0694	17.17	6.81×10^{-3}	0.0379	5.56
2	0.5	0.8883	0.8943	3.581	0.0310	0.1762	5.68
2	1.0	0.7887	0.7508	0.986	0.1000	0.5645	5.65
3	0.1	1.0327	0.3545	9.61×10^8	1.34×10^{-10}	6.15×10^{-9}	45.84
3	0.25	0.9890	0.4409	3734	3.31×10^{-5}	7.91×10^{-4}	23.89
3	0.5	0.9206	0.5862	47.15	2.44×10^{-3}	0.0346	14.17
3	1.0	0.7984	0.6682	3.703	0.0270	0.2683	9.96

46 at $c = 0.1$, while the finite element values of $\lambda_1(c) e^{2/c}$ move only from 2.2 at $c = 0.3$ to 3.0 at $c = 0.1$. The true prefactor is therefore close to constant, as in the upper bound, and the $\sqrt{c}$ of the lower bound is not sharp on $\mathbb{H}^3$. Table 9 examines the collapse itself: it gives $\lambda_1(c)$ on $\mathbb{H}^2$ for small c and its ratio to the envelope $\sqrt{c} e^{-1/(2c)}$ of Theorem 3.6. The ratio has a minimum near $c = 0.2$ and stays between 0.55 and 0.75; below $c = 0.1$ it increases steadily as c decreases, almost linearly in c (0.598 at $c = 0.1$, 0.691 at $c = 0.05$, 0.744 at $c = 0.025$). This behaviour is consistent with the two sided bound $\lambda_1(c) \asymp \sqrt{c} e^{-1/(2c)}$ of Theorems 3.6 and 3.7.

Relaxation rates in higher dimension. In polar coordinates the forward process on $\mathbb{H}^n$ is $dr = ((n - 1) \coth r - cr) dt + \sqrt{2} dW_r$ with a Brownian motion on S^{n-1} run at speed $\sinh^{-2} r$. For Table 10 we integrate it by Euler–Maruyama in the radial coordinate, with negative proposals reflected at $r = 0$, and by a geodesic random walk on S^{n-1} with step variance $2h \sinh^{-2} r$ per tangent coordinate, which is exact for $n = 2$ and first order accurate in general. Step sizes are $h = 0.002$ for $c \leq 2$ and $h = 0.001$ for $c \geq 4$, with 200,000 particles and four independent replicates per entry; the table reports the replicate mean and standard deviation. The angular runs start from the radial equilibrium law with the direction concentrated near e_1 , and the rate is the least squares slope of $\log \mathbb{E} \theta_1(X_t)$ on the window $0.05 \leq \mathbb{E} \theta_1 \leq 0.2$. The radial runs start from the radial equilibrium law shifted outwards by 0.3, and the rate is the slope of $\log |\mathbb{E} r_t - \mathbb{E}_{\mu_c} r|$ on the window $[0.02, 0.15]$; starting all particles at one radius instead excites transient radial modes of the opposite sign and biases this slope low by up to 5% at large c . Entries marked n/a are cases in which the angular decay does not reach the fitting window within the horizon $T = 6$. The $\mathbb{H}^2$ angular rates are re-estimated under this protocol as a control and agree with the gaps of Table 5 to within 2%. On $\mathbb{H}^2$ the radial rates fitted to $\mathbb{E} r_t$ are low by 1% to 6%, and Table 11 locates the cause. Any observable that is an exact eigenfunction of $-\mathcal{L}_c$ decays as a single exponential for every initial law, $\mathbb{E} f(X_t) = e^{-\lambda t} \mathbb{E} f(X_0)$, so its fitted slope contains no transient and measures only the integrator’s bias and the Monte Carlo noise. We take g_0 and g_1 from the finite element eigenvectors of the radial and first harmonic sectors, record $\mathbb{E} g_0(r_t)$ and $\mathbb{E} g_1(r_t) \theta_1$ alongside $\mathbb{E} r_t$ and $\mathbb{E} \theta_1$ on the same

Table 9: Weak confinement on $\mathbb{H}^2$: the gap and its ratio to the envelope $\sqrt{c}e^{-1/(2c)}$ of Theorem 3.6.

c	$\lambda_1(c)$	ratio	c	$\lambda_1(c)$	ratio
0.025	2.42×10^{-10}	0.744	0.08	3.44×10^{-4}	0.629
0.03	7.34×10^{-9}	0.733	0.1	0.0013	0.598
0.04	5.31×10^{-7}	0.712	0.15	0.0077	0.557
0.05	7.01×10^{-6}	0.691	0.2	0.0202	0.550
0.06	3.94×10^{-5}	0.669	0.25	0.0379	0.560

Table 10: Simulated relaxation rates on $\mathbb{H}^n$ against the sector gaps of Proposition 3.3: the angular observable $\mathbb{E} \theta_1(X_t)$ should decay asymptotically at $\lambda_1(c)$ and the radial observable $\mathbb{E} r_t - \mathbb{E}_{\mu_c} r$ asymptotically at $\lambda_1(c) + c$, so the last column should equal 1. Rates are means $\pm$ standard deviations over four replicates of 200,000 particles.

n	c	$\lambda_1(c)$ (FEM)	angular rate	ratio	$\lambda_1(c) + c$	radial rate	(radial – angular)/ c
2	0.5	0.1762	n/a	n/a	0.6762	0.663 ± 0.019	n/a
2	1	0.5645	0.567 ± 0.008	1.005	1.5645	1.488 ± 0.038	0.921
2	2	1.4736	1.471 ± 0.056	0.998	3.4736	3.266 ± 0.071	0.897
2	4	3.4118	3.472 ± 0.075	1.018	7.4118	7.334 ± 0.121	0.965
3	1	0.2683	0.271 ± 0.015	1.008	1.2683	1.274 ± 0.051	1.003
3	2	1.0324	1.036 ± 0.015	1.004	3.0324	3.026 ± 0.013	0.995
3	4	2.8691	2.877 ± 0.056	1.003	6.8691	6.836 ± 0.142	0.990
3	8	6.7729	6.869 ± 0.084	1.014	14.7729	14.467 ± 0.211	0.950
4	1	0.1009	n/a	n/a	1.1009	1.122 ± 0.022	n/a
4	2	0.6811	0.684 ± 0.005	1.004	2.6811	2.654 ± 0.076	0.985
4	4	2.3751	2.402 ± 0.058	1.011	6.3751	6.312 ± 0.211	0.978
4	8	6.1949	6.286 ± 0.082	1.015	14.1949	13.918 ± 0.237	0.954

trajectories, and fit the eigen-observables on the window between 60% and 5% of their initial value. Two integrators are compared: the polar scheme above at two step sizes, and a geodesic Euler scheme on the hyperboloid, $x \leftarrow \exp_x(-cr \nabla r h + \sqrt{2h} \xi)$ with ξ a standard Gaussian in $T_x \mathbb{H}^2$, which has no singular drift at the origin. On $\mathbb{H}^2$ the eigen-observable ratios sit within 2% of 1 in 21 of the 22 entries and within 3.5% in the remaining one, with no trend in the step size and no difference between the integrators, while the raw $\mathbb{E} r_t$ ratios on the same trajectories fall to 0.95 at $c = 2$. The same protocol on $\mathbb{H}^3$ and $\mathbb{H}^4$ gives ratios within 1.5% in all 16 entries, while the raw $\mathbb{E} r_t$ ratio on the same trajectories is 0.95 at $c = 8$. The deviation in Table 10 is therefore a property of fitting a multi-exponential signal on a fixed window, and the integrators are accurate at the percent level down to the origin. With the eigen-observables, the difference of the radial and angular rates divided by c lies between 0.97 and 1.02 in all 19 cases, against the value 1 of Proposition 3.3(iii).

Conditioning protocol. For Table 12, points at radius r are computed in double precision at 512 angles and rounded to each precision. The constraint residual $|\langle x, x \rangle_L + 1|$ is evaluated in that precision and maximised over those angles. Without reprojection, 400 geodesic Euler–Maruyama steps at $c = 2$ and $h = 0.01$ started from μ_2 let the residual grow to 2.6×10^{-4} in single and 5.7×10^{-13} in double precision. Changing chart does not escape the exponential: in the Poincaré ball a relative coordinate perturbation of size ε produces a geodesic error of order εe^r through the conformal factor.

C.2 Hierarchical generation (Section 5.2)

Data. Hierarchy A: groups $k \in \{0, 1, 2, 3\}$ at angles $k\pi/2$ with weights $(0.4, 0.3, 0.2, 0.1)$; two leaves per group at angular offsets ± 0.175 ; leaf centres $\exp_o(3.5(\cos \vartheta, \sin \vartheta, 0))$; samples $\exp_{\text{centre}}(0.15 \xi)$ with ξ a standard tangent Gaussian. Hierarchy B has three groups with the same leaf noise. A group of weight 0.6 at angle 0 and radius 4.5 has two subgroups at offsets ± 0.30 , each

Table 11: Finite element eigenfunction observables. Ratios of fitted rate to finite element eigenvalue for the eigen-observables $g_0(r_t)$ and $g_1(r_t)\theta_1$ (mean $\pm$ standard deviation over three replicates of 60,000 particles) next to the ratios for the raw observables $\mathbb{E} r_t$ and $\mathbb{E} \theta_1$ fitted on the same trajectories, for two integrators and two step sizes h on $\mathbb{H}^2$ and for the polar integrator on $\mathbb{H}^3$ and $\mathbb{H}^4$. The last column is the eigen-observable test of Proposition 3.3(iii), $(\text{radial rate} - \text{angular rate})/c$, which should equal 1. Entries marked n/a did not reach the fitting window within the horizon.

n	c	integrator	h	radial, eigen g_0	radial, raw $\mathbb{E} r_t$	angular, eigen $g_1\theta_1$	angular, raw $\mathbb{E} \theta_1$	(radial $-$ angular)/ c
2	0.5	polar	0.002	1.006 ± 0.030	0.971	1.010 ± 0.017	n/a	1.005
2	0.5	polar	0.001	0.982 ± 0.017	0.967	1.005 ± 0.033	n/a	0.974
2	0.5	hyperboloid	0.002	0.995 ± 0.029	0.985	0.988 ± 0.014	n/a	0.997
2	0.5	hyperboloid	0.001	0.981 ± 0.033	0.981	0.996 ± 0.014	n/a	0.976
2	1	polar	0.002	0.991 ± 0.003	0.996	1.020 ± 0.019	1.036	0.975
2	1	polar	0.001	1.019 ± 0.007	1.022	1.035 ± 0.009	1.060	1.010
2	1	hyperboloid	0.002	0.998 ± 0.012	0.981	0.993 ± 0.023	0.996	1.001
2	1	hyperboloid	0.001	0.988 ± 0.005	1.004	0.990 ± 0.055	0.985	0.987
2	2	polar	0.002	0.999 ± 0.010	0.948	1.012 ± 0.042	1.014	0.989
2	2	polar	0.001	0.998 ± 0.005	0.966	0.994 ± 0.002	0.992	1.001
2	2	hyperboloid	0.002	1.007 ± 0.005	0.985	0.995 ± 0.032	1.011	1.016
3	1	polar	0.002	1.003 ± 0.012	0.984	1.015 ± 0.008	1.103	1.000
3	2	polar	0.002	1.000 ± 0.009	0.989	1.002 ± 0.016	1.002	0.999
3	4	polar	0.001	0.998 ± 0.003	0.977	1.015 ± 0.013	1.014	0.986
3	8	polar	0.001	1.003 ± 0.004	0.947	0.999 ± 0.019	0.995	1.006
4	1	polar	0.002	1.008 ± 0.012	1.036	1.008 ± 0.018	n/a	1.008
4	2	polar	0.002	0.991 ± 0.009	0.998	0.998 ± 0.010	0.987	0.989
4	4	polar	0.001	1.001 ± 0.005	0.983	1.007 ± 0.036	1.009	0.997
4	8	polar	0.001	1.003 ± 0.002	0.953	1.005 ± 0.008	0.996	1.001

Table 12: Hyperboloid constraint error $\max |\langle x, x \rangle_L + 1|$ for points constructed at radius r (512 angles). The representation fails near $r \approx \frac{1}{2} \log(2/\varepsilon_{\text{mach}})$: about 8 in single precision and 18 in double.

r	2	5	8	12	16	20	24
fp32	1.8×10^{-6}	9.2×10^{-4}	0.36	1.3×10^3	3.8×10^6	8.6×10^9	2.4×10^{13}
fp64	5.3×10^{-15}	3.6×10^{-12}	4.7×10^{-10}	1.9×10^{-6}	7.8×10^{-3}	1.7×10^1	6.6×10^4

with two leaves at offsets ± 0.10 (four leaves of weight 0.15). A group of weight 0.3 at angle $2\pi/3$ and radius 2.5 has three leaves at offsets $-0.35, 0, 0.35$ (weight 0.1 each). A group of weight 0.1 at angle $4\pi/3$ and radius 3.5 has one leaf. Hierarchy C: a tree with branching (4, 4, 4). Each node's children lie at geodesic distance 1.3 from it, spread evenly within a cone of half angle $\pi/3$ about the direction away from the parent (the root's children over the full circle), after the construction of [13]. The 64 leaves lie at radii 3.53 to 3.88, nearest leaves 0.63 apart, with Dirichlet(2) weights spanning a factor 38 and the same leaf noise. Hierarchy W: the 1170 synsets of the WordNet subtree rooted at *mammal.n.01* (879 leaves, depth up to 9, 6448 pairs in the transitive closure of hypernymy). They are embedded in the Poincaré disk by the method of [12] as implemented in gensim (dimension 2, 10 negatives, 10 burn-in and 300 training epochs; `make_wordnet_embedding.py`). The trained embedding places most synsets at hyperbolic radius 8 to 12, so its hyperbolic radius is halved before use. This keeps the angular arrangement and brings nine tenths of the synsets to radius 3.9 to 6.1, with a median nearest neighbour distance of 0.40. All 1170 synsets are cluster centres with equal weight and noise 0.1. Group TV is computed over 39 groups: the 32 families at depth two (carnivore, ungulate, rodent, primate and so on, 14 of them single synsets) and, each as a group of its own, the root and the six synsets at depth one. On $\mathbb{H}^3$: group directions at the vertices $(\pm 1, \pm 1, \pm 1)/\sqrt{3}$ with an even number of minus signs, weights (0.4, 0.3, 0.2, 0.1), leaves at $\cos(\pm 0.175) \vartheta_g + \sin(\pm 0.175) \tau_g$ for a fixed unit tangent τ_g at each vertex, radius 3.5, the same noise. Hierarchies A, B and the $\mathbb{H}^3$ data have 8,192 training points, hierarchies C and W 32,768; all have 4,096 test points.

Networks and features. All models use a tanh multilayer perceptron with two hidden layers of 128 units, trained with Adam at learning rate 10^{-3} , reduced to 3×10^{-4} after 20k of 30k iterations,

and batch size 512. The time embedding contains t/T together with $\sin(2\pi kt/T)$ and $\cos(2\pi kt/T)$ for $k = 1, \dots, 4$, giving 9 features. The intrinsic model takes $\min(r, 12)$, angular Fourier features $\cos k\varphi, \sin k\varphi$ for $k \in \{1, 2, 4, 8\}$ and the time embedding, and outputs the frame coefficients (a_θ, b_θ) from Section 4.2. The tangent baseline takes $y \in \mathbb{R}^2, |y|$, the same Fourier features of $y/|y|$ and the same time embedding, ensuring that both models receive the same feature family. On $\mathbb{H}^3$ the intrinsic inputs are $\min(r, 12)$, the direction $\vartheta \in S^2$, its six products $\vartheta_i \vartheta_j$ ($i \leq j$) and the time embedding (19 features). The four outputs are a radial coefficient a_θ and a vector $w_\theta \in \mathbb{R}^3$. The residual is $u_\theta = a_\theta e_r + (w_\theta - (w_\theta \cdot \vartheta)\vartheta, 0)$. Its second term is the projection of w_θ onto the tangent space of S^2 at the direction ϑ of x , placed in the tangential part of $T_x \mathbb{H}^3$. Hence $|u_\theta|_g^2 = a_\theta^2 + |w_\theta - (w_\theta \cdot \vartheta)\vartheta|^2$ is bounded by the network output at every radius. The baseline’s inputs are $y \in \mathbb{R}^3, |y|$, the direction $y/|y|$ with its six products and the time embedding (22 features). For hierarchies C and W the width is 256 and the budget 40k iterations, with the learning rate reduced after 26.8k. The radial features are $\min(r, 12)$ and $\sin, \cos(\pi jr/6)$ for $j = 1, \dots, 4$, and the angular Fourier features use the doublings $k = 1, 2, 4, \dots$ up to $k = 64$ (C) and $k = 512$ (W). The baseline receives the same features of $y/|y|$ and $|y|$. The divergence threshold is $r > 7$ for C and $r > 8.1$ for W, about three and two units beyond the largest cluster radius.

Intrinsic training. We simulate a bank of forward trajectories once from the training set, with $K = 400$ steps, $h = 0.01$ and $T = 4$. For each batch element, we draw a start index k and a span $m \in \{1, 2, 4, 8\}$ uniformly, then use the target score $\hat{s} = \log_{x_{k+m}}(\Phi_{mh}(x_k))/(2mh)$ at x_{k+m} . Here Φ_τ is the geodesic flow of the drift alone, computed by m integrator steps, and the residual target is $u^* = \hat{s} + cr e_r$. The loss is the batch mean of $2\tau |u_\theta - u^*|_g^2$ with $\tau = mh$; the factor 2τ scales the target to unit variance across spans, so that no span dominates the objective. The span trades bias against variance. The drift flow centre $\Phi_\tau(x_k)$ approximates the conditional mean to first order in τ , which makes this part of the target error of order mh . In exchange, increasing m reduces the variance of the target by a factor of order m , a signal to noise gain of order $\sqrt{m}$; the largest span gives $mh = 0.08$. We have not derived the order of the full conditional score error on the manifold, which also carries the volume factor of the heat kernel.

Baseline training and sampling. The baseline is a variance preserving OU diffusion on $y = \log_o x \in \mathbb{R}^2$ with exact Gaussian conditionals, trained by exact denoising score matching using ε prediction at the same budget. Sampling uses 400 reverse Euler–Maruyama steps in the tangent plane followed by $\exp_o$, whereas the intrinsic sampler uses 400 reverse geodesic Euler–Maruyama steps of (2) with the anchored score. Its initial radius comes from numerical inverse transform of the exact radial density of μ_c ; the released code evaluates the CDF on a fine grid over $[0, 14]$ ($[0, 16]$ on $\mathbb{H}^3$) and interpolates it. The sampler caps the geodesic step norm at 0.5 and the drift norm at 50, with a numerical wall at $r = 30$. In the final models on hierarchy A the drift cap never engages, the wall is never reached and the step cap clips 0.21 to 0.23% of steps, close to the Gaussian tail probability $\mathbb{P}(\sqrt{2h}|\xi|_2 > 0.5) \approx e^{-6.25}$. The caps were chosen for data at radius 3.5 and do not suit much larger radii at strong confinement. At radius 10 and $c = 4$ the stationary drift has magnitude $cr = 40$, and the drift plus noise proposal would exceed the step cap of 0.5 on about 29% of the steps. For this reason hierarchy W is used at half its embedded radius in the sweep.

Metrics. MMD^2 is computed with the standard U-statistic formula and the Laplacian kernel $k(x, y) = e^{-d(x, y)/\sigma}$ on hyperbolic distance, with σ the median pairwise distance of 512 test points. Since the bandwidth is selected from the evaluation data, the usual unbiasedness guarantee for a fixed kernel does not strictly apply, and we do not call the estimate unbiased. The choice of kernel matters on a curved space. The Laplacian kernel is positive definite on $\mathbb{H}^n$ because the hyperbolic distance is conditionally negative definite, whereas the Gaussian kernel of geodesic distance is positive definite for every bandwidth only on flat spaces [35]. The submitted version used the Gaussian kernel $e^{-d^2/(2\sigma^2)}$ with the median heuristic, and on the test data of hierarchy A its Gram matrix has 55 eigenvalues below -10^{-8} , down to -0.7 . Under that kernel the prior μ_2 scores closer to the data than the data to itself (-7.5×10^{-3} against -0.05×10^{-3}), and every MMD^2 value in this version has therefore been recomputed with the Laplacian kernel. The estimator can be slightly negative, and

we report those values as computed. The other metrics count clusters and failures. Leaf total variation assigns each generated point to its nearest true leaf centre in hyperbolic distance (eight leaves on A, B and $\mathbb{H}^3$, 64 on C, 1170 synsets on W) and compares the resulting histogram with the true leaf weights. On W the group TV does the same over the 39 groups described above. Divergence is the fraction of samples with $r > 6$ or with non-finite coordinates; the data of A lie at $r \approx 3.5$, and the thresholds for C and W are given above. Every model is trained with three seeds, five on C and W, and each trained network is sampled with three seeds. A training seed fixes the network initialisation, the minibatch stream and the bank of forward trajectories, and it is the replication unit, since the three sample sets of one network share its weights and its bank. Tables 1, 14, 15 and 16 therefore report the mean and standard deviation over the training seed means, each the mean of three sample sets; for MMD^2 the last three tables add, in parentheses, the standard deviation over all nine (fifteen) sample sets. The prior mismatch is reported as the mean and standard deviation over three forward banks, those of the first three training seeds (`prior_mismatch_seeds.py`); on C and W these banks are simulated from the first 8,192 training points. The forward diagnostics use the angle φ of each trajectory endpoint on the hyperboloid: $R_4(T) = |\mathbb{E} e^{4i\varphi_T}|$ over the 8,192 training trajectories, and the prior mismatch is the MMD^2 estimate between the first 4,096 endpoints and an independent sample of μ_c of the same size, with the same kernel as above. Total compute for all generative runs is about twenty CPU hours.

D Full ablation and diagnostic results

Parameterisation ladder. Table 13 compares the three score heads of Section 4.2 at $c = 2.0$ on hierarchy A as a function of the training budget, with everything else fixed (`ablation_budget.py`). All three heads use the same data and bank of forward trajectories, width 128, single span targets, the bounded inputs $(\min(r, 12), \cos \varphi, \sin \varphi)$ and the time embedding. The run uses one training seed, one sampling seed and the Laplacian kernel; the percentages describe this run and are not a replicated estimate of a divergence probability. The heads differ only in the anchor and in how the network output becomes a tangent vector. The extrinsic heads output $w \in \mathbb{R}^3$ and project it with P_x , the anchored one adding $-\nabla W_c$ and projecting only the residual, whereas the frame head predicts two coefficients multiplying (e_r, e_φ) . With an untrained network the two extrinsic heads send 99% and 96% of the samples beyond $r = 6$ and the frame head none. The mechanism is the feedback loop of Section 4.2: once a sample leaves the data, a Euclidean output of size $O(1)$ becomes a tangent vector of size $\Theta(e^r)$ by Lemma B.1, and the next step moves the sample further out. Training removes the extrinsic divergence quickly: 1.4% and 0% at 500 iterations, 0.4% and 0.2% at 2000 and none at 6000. At 6000 iterations the three heads reach MMD^2 0.039, 0.039 and 0.045 and leaf TV 0.038, 0.057 and 0.085 with the capped sampler. Without the caps the untrained and lightly trained extrinsic heads produce non-finite coordinates, and the trained ones give the same values. The submitted version reported 28.7% and 27.1% divergence for the two extrinsic heads after 4k and 6k iterations with an earlier implementation; the released code does not reproduce those rates, and the table reports what it gives. The frame representation is therefore a structural guarantee against this mechanism, a head whose output cannot be amplified by the projection, rather than an accuracy advantage at a fixed budget. The accuracy of the full model comes from the Fourier features, the multi-span targets and the budget of Section 5.2 (Table 1).

Forward diagnostics. At $T = 4$, the $c = 2.0$ process has $R_4(T) = 0.010$ and $\mathbb{E}r(T) = 0.975$, consistent with the stationary mean radius 0.96 of μ_2 . At $c = 0.25$, $R_4(T) = 0.380$ compared with the data value 0.765, while the mean radius $\mathbb{E}r(T) = 3.91$ has moved from the initial 3.5 most of the way to the equilibrium mean 4.19. The forward process has therefore relaxed radially far more than angularly, the pattern described by Theorem 3.6 and quantified by Proposition 3.3: at $c = 0.25$ the radial rate $\lambda^{(0)} = 0.29$ gives a relaxation time close to the horizon, whereas the angular gap 0.038 gives a time scale of about 26. The prior mismatch measures how far the forward law still is from μ_c . With the Laplacian kernel and three forward banks per confinement (`prior_mismatch_seeds.py`), it is 13.1 ± 1.2 , 4.5 ± 1.3 , 0.3 ± 0.2 , 0.0 ± 0.0 and 0.1 ± 0.1 ($\times 10^{-3}$) on hierarchy A at $c = 0.25, 0.5, 1, 2$ and 4. On hierarchy B it is 26.0 ± 2.3 , 13.0 ± 1.3 ,

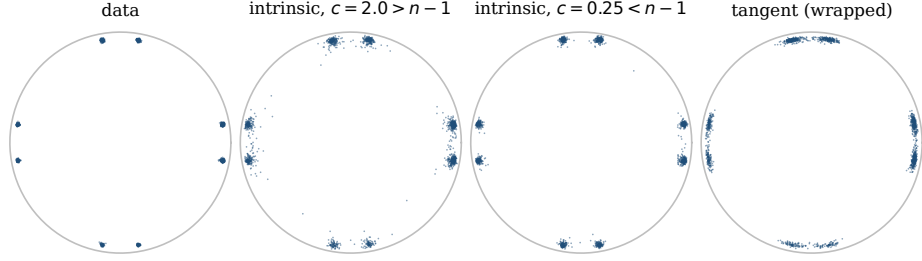


Figure 2: Data and generated samples in the Poincaré disk. Above threshold ($c = 2.0$) the intrinsic model keeps the group weights of the eight leaves and spreads each leaf more widely than the data. Below threshold ($c = 0.25$) the same architecture, budget and estimator keep the leaves tighter and flatten the group weights towards uniform (0.29, 0.28, 0.22 and 0.20 against 0.4, 0.3, 0.2 and 0.1). This happens because the forward process never destroyed the angular structure that the reverse process must reallocate. The tangent baseline resolves the leaves with the tangential elongation of the $\log_o$ chart.

Table 13: Ladder of score parameterisations at $c = 2.0$ on hierarchy A against the training budget: single span targets, angular features $\cos \varphi$ and $\sin \varphi$ only, one training and one sampling seed, the same data, forward bank, width and sampler for the three heads (ablation_budget.py). MMD^2 ($\times 10^3$, Laplacian kernel) and leaf TV are computed with the sampler’s caps. The diverged column gives the percentage of samples beyond $r = 6$ or non-finite with the caps and, after the slash, without them. Iteration 0 is the untrained network.

iterations	extrinsic, unanchored			extrinsic, anchored			frame, anchored		
	MMD ²	leaf TV	diverged	MMD ²	leaf TV	diverged	MMD ²	leaf TV	diverged
0	447.8	0.585	98.8/99.5%	425.4	0.637	96.2/96.4%	139.0	0.114	0.0/0.0%
500	59.8	0.200	1.4/2.2%	61.4	0.204	0.0/0.4%	48.0	0.099	0.0/0.0%
2000	49.7	0.125	0.4/0.5%	50.3	0.135	0.2/1.5%	48.1	0.098	0.0/0.0%
6000	38.7	0.038	0.0/0.0%	39.2	0.057	0.0/0.0%	45.0	0.085	0.0/0.0%

1.2 ± 0.6 , 0.0 ± 0.0 and 0.0 ± 0.0 ($\times 10^{-3}$) at the same confinements. On $\mathbb{H}^3$ it is 15.7 ± 0.2 , 1.6 ± 0.2 , 0.0 ± 0.1 , 0.0 ± 0.0 and 0.1 ± 0.0 ($\times 10^{-3}$) at $c = 0.5, 1, 2, 4$ and 8 (Tables 14 and 15; C and W in Table 16). The bank to bank standard deviation is at most about a tenth of the mean at the smallest confinement and grows relative to the mean as the mismatch approaches estimator noise. The diagnostic is therefore stable as a trend, and if it serves as a model selection statistic, a mean over a few banks should be used. Its decay can also be compared with the theory. Since the mismatch is a squared distance between mean embeddings, it decays on the scale $e^{-2\lambda_1(c)T}$ rather than $e^{-\lambda_1(c)T}$ when the first spectral mode dominates. Over the confinements with a mean mismatch above 10^{-4} , the slope of $\log$ mismatch against $\lambda_1(c)T$ is -1.8 on A, -1.5 on B, -2.4 on $\mathbb{H}^3$, -3.0 on C and -1.6 on W, around the value -2 of a single mode. With two or three points per hierarchy and several modes present, this is a consistency check, not a fit. On A, B and W the point at $c = 1$ is within two standard deviations of zero, and without it the slopes are -1.9 , -1.3 and -0.9 . The diagnostic uses only the forward process; if the mismatch remains appreciable at the chosen horizon, the reverse sampler is initialised from a law that the forward process has not reached.

Training variability. Table 14 repeats Table 1 for hierarchies A and B with the standard deviation over all nine sample sets in parentheses, and Figure 3 plots the sweep on every hierarchy. The training seed is the replication unit, because the three sample sets of one network share its weights and its bank of trajectories. The deviation of MMD^2 over all nine sample sets is between 0.9 and 1.6 times the training seed deviation (at $c = 2$ on hierarchy A, 0.63×10^{-3} against 0.57×10^{-3}). With three training seeds the deviations have two degrees of freedom and are descriptive. The orderings reported in Section 5.2 nevertheless hold seed by seed. On hierarchy A the MMD^2 of the three training seeds is 2.0, 2.4 and 2.3 at $c = 1$, against 2.7, 2.7 and 3.7 at $c = 2$, 6.0, 7.4 and 6.4 at $c = 4$ and 8.5, 9.9

and 8.1 at $c = 0.25$ (all $\times 10^{-3}$). In every seed $c = 1$ is below $c = 2$, which is below $c = 4$, with $c = 0.25$ worst. On hierarchy B, by contrast, the models at $c = 1$ and $c = 2$ are not separated by the training seeds (8.0, 7.1, 5.9 against 7.5, 6.9, 8.2). The leaf TV of the wrapped baseline varies more across training seeds than across sampling seeds on hierarchies A and B. No sample diverged on hierarchy A; on hierarchy B at most 0.07% of samples diverged in any run. These uncertainties are standard deviations over seeds and not formal significance tests.

Table 14: Hierarchical generation on $\mathbb{H}^2$: mean $\pm$ standard deviation over the three training seeds (each the mean of three sample sets), with the standard deviation over all nine sample sets in parentheses. The mismatch is the mean $\pm$ standard deviation over three forward banks. MMD^2 and mismatch are multiplied by 10^3 ; lower is better. When the first spectral mode dominates, MMD^2 decays on the scale $e^{-2\lambda_1(c)T}$, since it is a squared distance between mean embeddings. The mismatch is consistent with this scaling. The generation error is smallest at the threshold and larger on both sides.

model	$\lambda_1(c)T$	hierarchy A			hierarchy B		
		$\text{MMD}^2 (\times 10^3)$	leaf TV	mismatch	$\text{MMD}^2 (\times 10^3)$	leaf TV	mismatch
intrinsic, $c = 0.25$	0.15	8.8 ± 1.0 (1.3)	0.133 ± 0.008	13.1 ± 1.2	35.1 ± 1.6 (2.4)	0.239 ± 0.010	26.0 ± 2.3
intrinsic, $c = 0.5$	0.70	3.3 ± 0.5 (0.7)	0.072 ± 0.010	4.5 ± 1.3	17.6 ± 1.5 (1.4)	0.155 ± 0.009	13.0 ± 1.3
intrinsic, $c = 1.0$	2.26	2.2 ± 0.2 (0.2)	0.040 ± 0.003	0.3 ± 0.2	7.0 ± 1.1 (1.0)	0.065 ± 0.012	1.2 ± 0.6
intrinsic, $c = 2.0$	5.89	3.0 ± 0.6 (0.6)	0.047 ± 0.007	0.0 ± 0.0	7.6 ± 0.6 (0.8)	0.053 ± 0.009	0.0 ± 0.0
intrinsic, $c = 4.0$	13.65	6.6 ± 0.7 (0.7)	0.063 ± 0.012	0.1 ± 0.1	16.8 ± 0.9 (0.9)	0.103 ± 0.019	0.0 ± 0.0
wrapped baseline	–	5.6 ± 0.3 (0.4)	0.039 ± 0.013	–	13.1 ± 0.5 (0.5)	0.032 ± 0.012	–

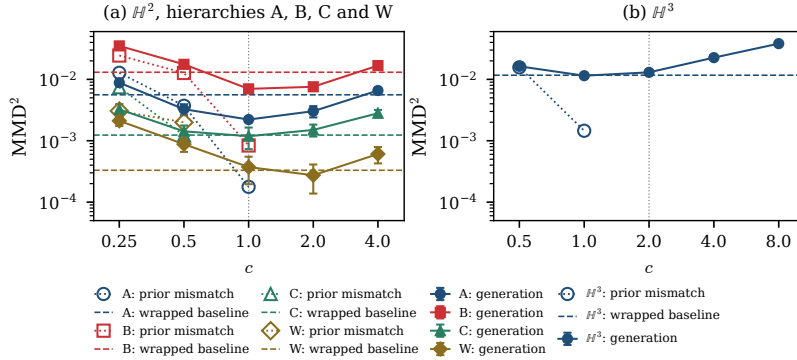


Figure 3: Generation error against confinement on (a) $\mathbb{H}^2$, hierarchies A, B, C and W, and (b) $\mathbb{H}^3$. Filled markers: MMD^2 between generated samples and test data, mean $\pm$ standard deviation over nine sample sets (fifteen on C and W). Open markers: the prior mismatch, the MMD^2 between the forward endpoint cloud of one bank and an exact sample of μ_c , which needs no network (values zero within estimator noise omitted). Dashed lines: the wrapped baselines; dotted vertical line: the threshold $c = n - 1$.

Hierarchy B. Figure 4 shows hierarchy B in the Poincaré disk for the data, the intrinsic model at $c = 2$ and $c = 0.25$ and the baseline, with the conventions of Figure 2. The heavy group at radius 4.5 separates the models. The intrinsic model at $c = 2$ keeps the group weights (0.58, 0.30 and 0.12 in the plotted samples against 0.6, 0.3 and 0.1) while spreading the four heavy leaves along the boundary, with a median distance to the nearest leaf of 0.75 against 0.18 in the data. At $c = 0.25$ the trade is reversed: these leaves are more compact, 0.48, whereas the group weights are flattened to 0.37, 0.44 and 0.19. The baseline stretches the heavy leaves into a long arc along the boundary, 1.44, the tangential elongation of the $\log_o$ chart.

The hierarchy on $\mathbb{H}^3$. The hierarchy places four groups at the vertices of a tetrahedron, with the weights, leaf offsets, radius and noise of hierarchy A (Appendix C). The intrinsic model uses the

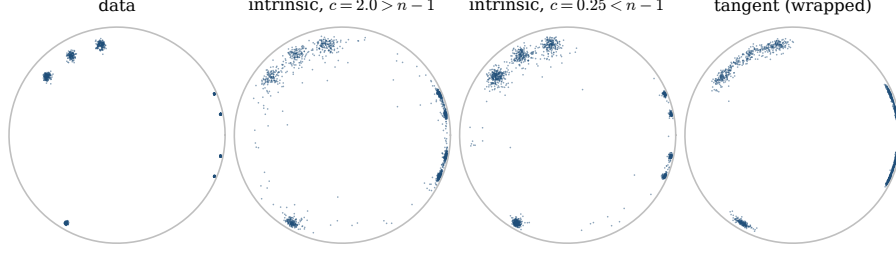


Figure 4: Hierarchy B: data and generated samples in the Poincaré disk, conventions as in Figure 2.

radial scalar head and the projected tangential vector head of Section 4.2 (Appendix C, Networks and features) with inputs $\min(r, 12)$, the direction $\vartheta \in S^2$ with its degree two products and the time embedding. The baseline is an OU diffusion in $\mathbb{R}^3$ with the same features. The threshold is $c = 2$, and we train at $c \in \{0.5, 1, 2, 4, 8\}$ with the seeds, budgets and metrics of Section 5.2. Table 15 gives the sweep. The ordering $c = 1$ below $c = 2$ below $c = 0.5$ below $c = 4$ below $c = 8$ holds for every training seed, and leaf TV is smallest at the threshold. Figure 5 shows the direction $\vartheta \in S^2$ of generated points in longitude and latitude, coloured by radius. The data leaves are tight in direction, since noise of scale 0.15 at radius 3.5 subtends an angle of about 0.01. At $c = 1$ the eight leaves are resolved with a modest angular spread. At $c = 4$ the spread within each leaf is visibly larger and stray points appear between groups, the visual counterpart of the larger MMD^2 . The baseline has the fewest stray points, with a slightly wider spread within each leaf than at $c = 1$.

Table 15: Hierarchical generation on $\mathbb{H}^3$, threshold $c = 2$; conventions as in Table 14. No sample diverged in any run.

model	$\lambda_1(c)T$	$\text{MMD}^2 (\times 10^3)$	leaf TV	mismatch
intrinsic, $c = 0.5$	0.14	16.3 ± 1.3 (1.5)	0.133 ± 0.013	15.7 ± 0.2
intrinsic, $c = 1.0$	1.07	11.5 ± 1.0 (0.9)	0.079 ± 0.014	1.6 ± 0.2
intrinsic, $c = 2.0$	4.13	13.0 ± 0.3 (0.4)	0.059 ± 0.009	0.0 ± 0.1
intrinsic, $c = 4.0$	11.48	22.6 ± 2.2 (2.0)	0.079 ± 0.028	0.0 ± 0.0
intrinsic, $c = 8.0$	27.09	38.2 ± 1.6 (1.5)	0.105 ± 0.013	0.1 ± 0.0
wrapped baseline	—	11.7 ± 1.0 (0.9)	0.043 ± 0.003	—

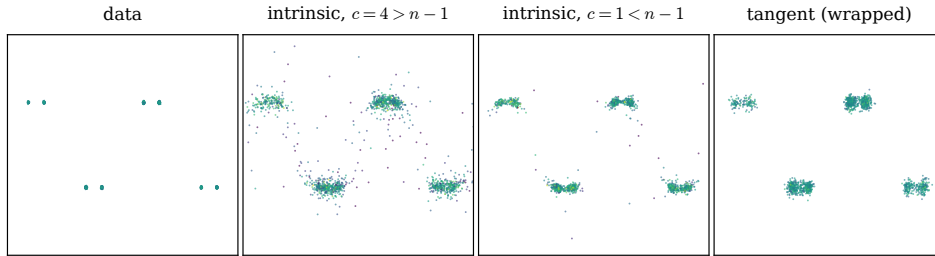


Figure 5: Hierarchy on $\mathbb{H}^3$: directions of data and generated samples in longitude and latitude, colour giving the radius (viridis, 2.5 to 4.5).

Hierarchies C and W. Table 16 gives the sweep on the two larger hierarchies, with the conventions of Table 14; the floor rows give the values of the test set itself against the training set, set by 4,096 samples over 64 and 1170 clusters. At most 0.02% of samples diverged in any run on C and 0.15% on W. Figures 6 and 7 show the data and samples in the chart $\log_o$, geodesic polar coordinates $(r \cos \varphi, r \sin \varphi)$, together with one sector in the coordinates (φ, r) ; unlike the Poincaré disk, this chart does not compress the radius. On both hierarchies the difficulty is the fine structure. The

nearest clusters are 0.63 apart on C and a median 0.40 on W in hyperbolic distance, against 3.6 on hierarchy A, and the baseline therefore faces the same fine structure as the intrinsic model. The angular Fourier features of the networks sit at the doublings $k = 1, 2, \dots, 64$ (C) and up to 512 (W). Three diagnostics on C, each for the first training seed at $c = 1$ and for the baseline with everything else fixed, test whether the unresolved leaves are a matter of the configuration (Figure 8 shows two of them; `experiments_tree_dense.py`, `experiments_tree_big.py`, `experiments_tree_fine.py`). The standard configuration gives MMD^2 0.72×10^{-3} and leaf TV 0.160 at $c = 1$, and 0.82 and 0.227 for the baseline. The dense basis $\cos k\varphi, \sin k\varphi$ for every $k \leq 64$ gives 1.04 and 0.121 at $c = 1$, and 0.59 and 0.218 for the baseline. Four times the budget, width 512 and 160k iterations, gives 1.53 and 0.136 at $c = 1$ and 1.39 and 0.227 for the baseline, with faint stripes in the sector plot. With this budget the MMD^2 is larger for both models, within the spread of the five training seeds of the standard configuration (0.72 to 1.60 at $c = 1$, 0.79 to 2.34 for the baseline). A finer discretisation, 1000 steps of $h = 0.004$ for the bank, the targets and the sampler, gives 1.00 and 0.145 at $c = 1$ and 0.65 and 0.218 for the baseline. The leaves within a branch remain unresolved in every case.

Hierarchy W at full radius. A final experiment uses the embedding at its full radius (synsets mostly at radius 8 to 12, nearest neighbours a median 5.8 apart). The sampler’s caps are raised to a drift of 200 and a step of 1.5, and the radial features extend to $r = 14$. One training seed per model gives the same ordering: MMD^2 0.49, 0.38, 0.88, 0.99 and 1.90 ($\times 10^{-3}$, kernel bandwidth 19) at $c = 1, 2, 4, 0.5$ and 0.25 against 0.34 for the baseline. Below the threshold the mismatch is 9.6 at $c = 0.25$ and 3.6 at $c = 0.5$ ($\times 10^{-3}$), and one sample diverged, at $c = 0.25$. The synset level remains unresolved: node TV is far from its floor for every model, 0.41 to 0.46 against 0.21. The baseline has the lowest MMD^2 , while its node TV 0.446 and group TV 0.072 are worse than the intrinsic model’s 0.412 and 0.057 at $c = 1$; with one seed per model these differences are descriptive (`experiments_wordnet_full.py`).

Table 16: Hierarchical generation on hierarchies C and W: mean $\pm$ standard deviation over the five training seeds (each the mean of three sample sets), with the standard deviation over all fifteen sample sets in parentheses. The mismatch is the mean $\pm$ standard deviation over three forward banks of 8,192 points. MMD^2 and mismatch are multiplied by 10^3 . The sampling floor is the test set itself against 4,096 training points.

model	$\lambda_1(c)T$	$\text{MMD}^2 (\times 10^3)$	leaf or node TV	group TV	mismatch
<i>hierarchy C (64 leaves at depth three)</i>					
intrinsic, $c = 0.25$	0.15	3.24 ± 0.63 (0.71)	0.171 ± 0.009	–	7.09 ± 0.53
intrinsic, $c = 0.5$	0.70	1.42 ± 0.31 (0.35)	0.147 ± 0.005	–	1.36 ± 0.28
intrinsic, $c = 1.0$	2.26	1.18 ± 0.41 (0.46)	0.160 ± 0.015	–	0.01 ± 0.05
intrinsic, $c = 2.0$	5.89	1.50 ± 0.27 (0.33)	0.187 ± 0.005	–	0.02 ± 0.01
intrinsic, $c = 4.0$	13.65	2.80 ± 0.30 (0.37)	0.239 ± 0.004	–	0.01 ± 0.04
wrapped baseline	–	1.24 ± 0.67 (0.66)	0.225 ± 0.008	–	–
sampling floor	–	0.18	0.054	–	–
<i>hierarchy W (1170 WordNet synsets, 39 groups)</i>					
intrinsic, $c = 0.25$	0.15	2.11 ± 0.34 (0.38)	0.284 ± 0.004	0.080 ± 0.008	3.22 ± 0.24
intrinsic, $c = 0.5$	0.70	0.88 ± 0.15 (0.22)	0.277 ± 0.004	0.049 ± 0.007	1.95 ± 0.07
intrinsic, $c = 1.0$	2.26	0.37 ± 0.15 (0.18)	0.280 ± 0.002	0.044 ± 0.008	0.13 ± 0.09
intrinsic, $c = 2.0$	5.89	0.27 ± 0.06 (0.14)	0.287 ± 0.004	0.047 ± 0.004	0.07 ± 0.06
intrinsic, $c = 4.0$	13.65	0.61 ± 0.12 (0.18)	0.298 ± 0.003	0.055 ± 0.004	0.01 ± 0.01
wrapped baseline	–	0.33 ± 0.03 (0.06)	0.313 ± 0.002	0.062 ± 0.004	–
sampling floor	–	0.05	0.214	0.027	–

Sampler caps and step size. The intrinsic sampler caps the geodesic step at 0.5 and the drift at 50, and the rise of the error above the threshold could in principle be an artefact of these caps or of the reverse step. To test this, we retrained the first training seed of every intrinsic model on hierarchy A and on $\mathbb{H}^3$. We then sampled each network with its three sampling seeds at the reverse step $h = 0.01$ of the paper and at $h = 0.005$ and $h = 0.0025$, counting the capped steps (`sampler_caps.py`; Table 17). The caps rarely engage. At $h = 0.01$ the step cap engages in 0.21 to 0.23% of the steps on

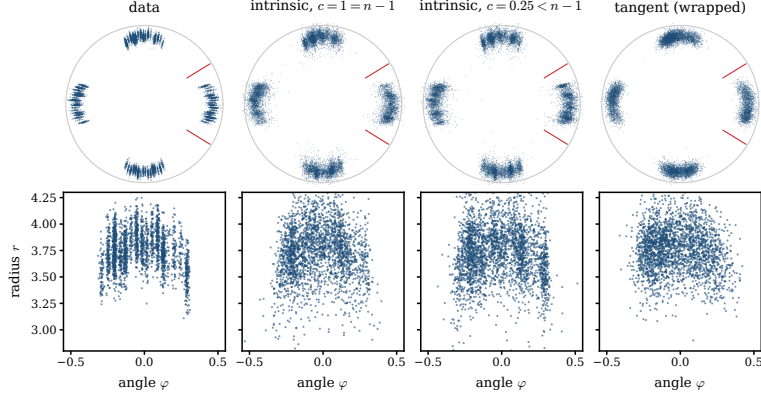


Figure 6: Hierarchy C: data and generated samples in the chart $\log_o$ (top) and in the sector marked in red, in the coordinates radius against angle (bottom), 12,288 points per panel. The four branches and their weights are reproduced; the sixteen leaves of a branch, visible as stripes in the data, are smeared into one cloud by every model.

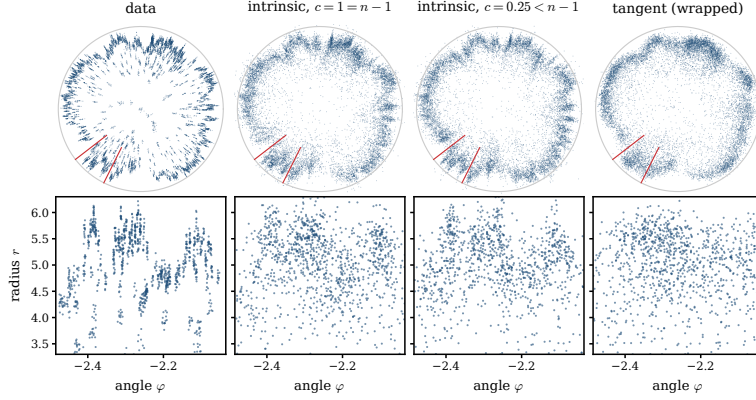


Figure 7: Hierarchy W, built on the WordNet mammal taxonomy: as Figure 6, with the sector containing the primates. The angular density of the families is reproduced at $c = 1$ and the synset level is not.

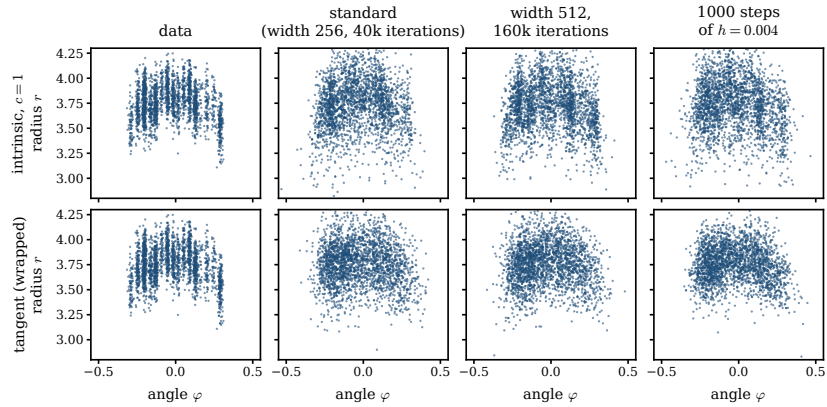


Figure 8: Hierarchy C, the sector of Figure 6: the data (left), then the intrinsic model at $c = 1$ (top) and the wrapped baseline (bottom) in the standard configuration, with four times the budget and with the finer discretisation; first training seed, 12,288 points per panel.

$\mathbb{H}^2$ and in 0.65 to 0.70% on $\mathbb{H}^3$, independently of c . These shares are close to the tail probability of the Gaussian increment alone, $\mathbb{P}(\sqrt{2h}|\xi| > 0.5)$ with ξ standard normal in two or three dimensions (0.19% and 0.59%). The drift cap never engages. At $h = 0.005$ at most 0.002% of the steps are capped, and at $h = 0.0025$ none. The step size does not change the shape of the sweep. With the finer steps the MMD^2 changes by $+0.02$ to -0.84×10^{-3} , at most 16%, on hierarchy A and by $+0.39$ to -1.84×10^{-3} , at most five percent, on $\mathbb{H}^3$. The minimum at $c = 1$ and the rise at large c are the same at all three step sizes; only $c = 0.5$ and $c = 2$ on hierarchy A, within 0.01×10^{-3} of each other at $h = 0.01$, change places. The rise at large c is therefore not produced by the caps or by the discretisation. This experiment does not determine whether the rise comes from the score estimate itself or from the sampler’s sensitivity to score error.

Table 17: Sampler caps and reverse step size, first training seed of each intrinsic model: fraction of reverse steps at which the geodesic step cap engaged (the drift cap never did) and MMD^2 ($\times 10^3$, mean $\pm$ standard deviation over three sampling seeds) at the reverse step $h = 0.01$ of the paper and at two finer steps, with the same trained network.

hierarchy	c	$h = 0.01, K = 400$		$h = 0.005, K = 800$		$h = 0.0025, K = 1600$	
		capped steps	MMD^2	capped steps	MMD^2	capped steps	MMD^2
A ($\mathbb{H}^2$)	0.25	0.22%	8.5 ± 1.1	$< 0.01\%$	8.1 ± 0.6	0.00%	8.2 ± 1.0
	0.5	0.21%	2.7 ± 0.3	$< 0.01\%$	2.7 ± 0.4	0.00%	2.7 ± 0.4
	1	0.22%	2.0 ± 0.1	$< 0.01\%$	1.8 ± 0.4	0.00%	1.7 ± 0.3
	2	0.21%	2.7 ± 0.4	$< 0.01\%$	2.5 ± 0.2	0.00%	2.3 ± 0.2
	4	0.23%	6.0 ± 0.2	$< 0.01\%$	5.8 ± 0.2	0.00%	5.2 ± 0.8
$\mathbb{H}^3$	0.5	0.65%	17.3 ± 1.0	$< 0.01\%$	17.2 ± 0.8	0.00%	17.1 ± 0.7
	1	0.65%	10.4 ± 0.4	$< 0.01\%$	10.3 ± 0.2	0.00%	10.2 ± 0.8
	2	0.65%	12.9 ± 0.1	$< 0.01\%$	13.3 ± 0.3	0.00%	12.7 ± 0.2
	4	0.67%	23.6 ± 1.0	$< 0.01\%$	22.9 ± 1.5	0.00%	22.7 ± 0.2
	8	0.70%	40.0 ± 0.5	$< 0.01\%$	38.2 ± 1.1	0.00%	38.2 ± 0.4

Code availability

All code, result files and verification scripts are available at
<https://github.com/dgodorozha/2026-NeurIPS-STODY-Camera-Ready>.

Disclosure of AI assistance

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